



Quantification of retinal microvascular imaging features from fundus photos in ocular and systemic disease: a framework for standardization[☆]

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ABSTRACT

Microvascular dysfunction is increasingly recognized as an important contributor to ocular and systemic diseases, including diabetic retinopathy, cardiovascular disease, and Alzheimer's disease and related dementias. Elucidating how early microvascular injury contributes to disease pathobiology, and development of sensitive, quantifiable features of microvascular dysfunction, represents a critical frontier for improving risk stratification, enabling earlier diagnosis and guiding targeted interventions. In this context, retinal imaging has emerged as a powerful modality, enabling non-invasive, high-resolution visualization of the microvasculature with color and (ultra)widefield fundus imaging. Rapid technological advances have led to an array of, open-source tools for quantitative extraction of retinal microvascular features. However, progress in this field is hampered by poor comparability between tools, limiting reproducibility and cross-study integration. Differences between tools originate from variations in image processing steps, including vessel segmentation, arteriole-venule classification, optic disc detection, region-of-interest definition, image quality assessment, and the algorithms for metric calculation. In this review we comprehensively compare existing analytical analysis tools for *static* vessel analyses and delineate how these methodological differences influence the vascular measure quantification. To advance robust, scalable biomarker development, we propose a methodological framework to standardize and harmonize retinal microvascular quantification. This framework can guide future studies in addressing key gaps in literature and in developing imaging biomarkers for clinical use. Critical open questions include *whether* and *when* retinal imaging features change during ocular and systemic disease, whether microvascular dysfunction is *reversible*, how microvascular dysfunction contributes to neurodegeneration, and how central and peripheral retinal microvascular dysfunction differ.

1. Introduction

Non-invasive imaging of the human body has advanced dramatically over the last century. The development of magnetic resonance imaging (MRI) and ultrasound have enabled high-resolution imaging of the brain, heart, large vessels, and abdominal organs such as the kidneys (Brock et al., 2023). Similarly, advances in ocular imaging have led to several transformative advances, including standard fundus (field of view 25°–60°) and (ultra)widefield (color) photography (field of view up to 200°) (Zhu et al., 2025a; Quinn et al., 2019).

The retina allows non-invasive, accurate, and inexpensive visualization of microvascular structures, enabling quantification of microvascular health from imaging features. With standard fundus photos and (ultra)widefield imaging, arterioles and venules with diameters between ~30 and 300 μm can be visualized from single static images (Panwar et al., 2016). As the retinal microvasculature is physiologically and anatomically similar to that in other organs (e.g., the brain, heart, and kidney), retinal microvascular imaging features are important tools for the study of microvascular health (sometimes termed *vasculometry*) in other organ systems; and may provide scalable biomarkers for ocular and systemic disease (Zhu et al., 2025a). Indeed, retinal metric measures have been found to be associated with parameters related to the brain (e.g. brain volume, signs of cerebral small vessel disease) (Tong et al., 2025; Wu et al., 2025), heart (cardiac structure (Huang et al., 2020), heart rate variability (Wentzel et al., 2020), and cardiac blood-based biomarkers (Wentzel et al., 2019)), and kidney (estimated glomerular filtration rate (Eriksen et al., 2015) and albuminuria (Paterson et al., 2021)).

Over the last two decades, research on retinal vasculometry has yielded important scientific insight into the contributions of microvascular dysfunction to ocular and systemic diseases (Zhu et al., 2025a; Cheung et al., 2012, 2015, 2022a; Ikram et al., 2013; Lee et al., 2025; Li et al., 2016a; Liew et al., 2008; Sasongko et al., 2010; Wong and McIntosh, 2005; Wong et al., 2001; Kee et al., 2017; Zhang et al., 2026). Indeed, retinal microvascular changes are associated with ocular diseases with a vascular origin (e.g. diabetic retinopathy and glaucoma)

and diseases that affect other organs systems such as cardiovascular disease (stroke, myocardial infarction, heart failure), chronic kidney disease, Alzheimer's disease and related dementias, Parkinson's disease, schizophrenia, bipolar disorder, and (late-life) depression (Zhu et al., 2025a; Cheung et al., 2012, 2015, 2022a; Ikram et al., 2013; Lee et al., 2025; Li et al., 2016a; Liew et al., 2008; Sasongko et al., 2010; Wong and McIntosh, 2005; Wong et al., 2001; Kee et al., 2017; Zhang et al., 2026; Cao et al., 2024; Dumitrascu et al., 2021, 2024; Gonzales et al., 2025; Shi et al., 2020, 2023; Antonetti et al., 2021; Ashraf et al., 2020). Many of these observations have been shown in large-scale population studies across the entire life span (Zhu et al., 2025a; Cheung et al., 2012, 2015, 2022a; Ikram et al., 2013; Lee et al., 2025; Li et al., 2016a; Liew et al., 2008; Sasongko et al., 2010; Wong and McIntosh, 2005; Wong et al., 2001; Kee et al., 2017; Zhang et al., 2026).

Despite these important observations and findings, critical gaps remain in our understanding of the role of retinal microvascular dysfunction in ocular and systemic diseases (Zhu et al., 2025a; Cheung et al., 2015, 2022a; Ikram et al., 2013; Lee et al., 2025). No retinal vasculometry features are currently used in clinical practice. We can identify various reasons: 1) few longitudinal studies have evaluated the *temporality* and *reversibility* of retinal microvascular imaging features (Lee et al., 2025; Chew et al., 2025). This precludes conclusions on whether retinal microvascular imaging features may be tools for individualized risk management, monitoring of disease progression, and response to therapies (Lee et al., 2025; Chew et al., 2025; Ahlbom, 2021). 2) few studies have investigated the retinal *peripheral* microvasculature, which may be more susceptible to deterioration than the central retinal microvasculature (Zhu et al., 2025a; Quinn et al., 2019; Cheung et al., 2015, 2022a; Ikram et al., 2013; Lee et al., 2025; Dumitrascu et al., 2021, 2024). 3) the contribution of microvascular dysfunction to core pathophysiology of neurodegeneration and dysfunction of the neurovascular coupling unit remains incompletely understood (van der Heide et al., 2022). 4) there are relatively few studies on the reproducibility of retinal imaging features especially across different populations (Eid et al., 2025; Mautuit et al., 2022; Yip et al., 2016; French and Heitmar, 2021).

Technological advances in image processing over the last 20 years have led to the availability of a large number of tools for quantifying static retinal microvascular features in standard color and increasingly

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(ultra)widefield fundus images (Mahapatra et al., 2024). In the earliest stages of research, semi-automatic tools were used to quantify retinal microvascular imaging features from the posterior pole (i.e., central retina) around the year 2000 (Liew et al., 2008). These tools require human input to check vessel segmentation, arteriole-venule classification and optic disc position detection, which is time-consuming (processing times between 10 and 20 min per image have been reported) (Shi et al., 2022). Since 2015 and the advent of artificial intelligence (AI) and deep learning, fully automated tools have become available that enable rapid quantification of central retinal microvascular imaging features (i.e., within seconds to minutes an image can be processed). The availability of fully automated tools makes it feasible for individual researchers to analyze larger sample sizes, which are increasingly available (e.g., UK Biobank). Most available tools are for the analysis of the central retinal microvasculature; there are fewer tools for the study of the peripheral retinal microvasculature (Ashraf et al., 2021; Csincsik et al., 2018; Leung et al., 2024; Pellegrini et al., 2018). Some of these tools are proprietary, some are commercially available, and only a small number of tools are open source (i.e., publicly available) (Vargas Quiros et al., 2025; Zhou et al., 2022; Fhima et al., 2023, 2024).

The increasing availability of fully automated AI tools for analysis and quantification of the retinal microvasculature provides new opportunities to understand the role of microvascular dysfunction in ocular and systemic diseases, and potentially for the development of retinal vasculometry-derived biomarkers for clinical use (Chew et al., 2025). However, a significant barrier is that the results obtained from different analytic tools compare poorly (Eid et al., 2025; Mautuit et al., 2022; Yip et al., 2016; French and Heitmar, 2021). For example, a comparison between two analysis tools showed that the correlation between most quantitative microvascular metrics determined from the same image was lower than 0.40 (Eid et al., 2025).

In view of the above, the main aim of this review is to comprehensively compare commonly used tools for quantifying central and peripheral retinal microvascular imaging features and propose a methodological framework for their standardization. The review is structured as follows. In the *Background* section, we briefly summarize key aspects of retinal microvasculature anatomy and physiology. In the *Current methods* section, we describe and compare commonly used tools for analysis of the central and peripheral retinal microvasculature. In the *Proposed framework for standardization* section, we present a methodological framework and recommendations for standardizing the quantification of retinal microvascular imaging features. Finally, in the *Discussion* section, we outline directions for future research.

2. Context

Microvascular dysfunction is considered a disease process that affects the retina and other organ systems throughout the body. To a certain extent ocular and systemic risk factors may have similar (detrimental) effects on the retinal microvasculature. Biologically, microvascular dysfunction is thought to result in impaired ability to optimally regulate metabolism and stabilize and regulate hydrostatic pressure at the capillary level. Microvascular dysfunction thus increases the risk of impaired metabolism and capillary hypertension and rupture, both of which predispose to local ischemia and subsequent tissue damage (e.g. signs of retinopathy in the retina and signs of cerebral small vessel disease in the brain).

Many systemic risk factors that are detrimental for the microvasculature may have similar effects, but the exact effects may differ between risk factors. For example, systemic factors such as hypoxia and hyperglycemia may have different effects on retinal arteriolar diameters than hypertension. The systemic context is thus relevant for the interpretation of vascular measures and selection of measures that are most informative on disease presence and progression. Hence, standardization of retinal microvascular imaging feature quantification is relevant both for the study of ocular and systemic diseases (such as

cardiovascular, renal, and neuropsychiatric diseases).

In this review, we focus on specific aspects of microvascular dysfunction that can be inferred from static fundus-derived imaging features. The biological changes underlying changes in retinal vasculometry may reflect (temporary) *functional changes* (e.g., arteriolar or venular widening) and/or *structural changes* (e.g., loss of collagen in the vessel wall) (Ikram et al., 2013; Hanssen et al., 2022). This review does not address quantitative retinal microvascular measures assessed with functional tests (e.g., change in retinal microvascular diameter during exposure to a light stimulus, as can be measured with the dynamic vessel analyzer) or microvascular changes from other modalities (e.g., optical coherence tomography angiography (OCT-A)) (Van Eijgen et al., 2025).

3. Section 1: Background

The human retina has an intricate vascular network that supplies oxygen and nutrients to support visual function. This network is not uniform across the retina. There is regional heterogeneity between the central and peripheral regions. This heterogeneity is likely due to the functional differences. The vessels in the central retina support high-acuity vision, whereas in the periphery, they support visual field awareness and motion detection. OCT-A and ultra-widefield imaging can now help in analyzing the details of these regional vascular differences (Lavia et al., 2020; Mastropasqua et al., 2019; Zhang et al., 2018). Understanding the anatomical, functional, and physiological distinctions between central and peripheral retinal vasculature is crucial for interpreting clinical findings, understanding disease pathophysiology, and developing region-specific therapeutic approaches. In this section, we provide an overview of key aspects of the anatomy and physiology of central and peripheral retinal microvasculature to provide a biological context before discussing the quantification of features.

3.1. Development of the retinal microvasculature

The development of retinal blood vessels is region-specific, which underlies the central-to-peripheral differences characteristic of the mature retina. During early development, astrocytes and vessels grow around, but not through, the foveal area, forming the foveal avascular zone (FAZ) (Provis et al., 1997). The superficial vascular network forms first, followed later by the deep capillary network, a sequence that likely contributes to regional variations in vascular organization and results from complex interactions between endothelial cells and astrocytes (Soetikno et al., 2015). The developmentally distinct peripheral vessels have different relationships with their supporting cells and with angiogenic signalling and metabolic demand. The intermediate capillary plexus is found only in the central retina and serves the high metabolic demand. The intermediate plexus is absent in the periphery.

3.2. Organization of the retinal microvasculature

The inner retina receives arterial blood supply from the central retinal artery and (if present) cilioretinal arteries.^{56, 57} The ophthalmic artery receives blood from the carotid artery and branches into several vessels, including 1) the central retinal artery and 2) the short posterior cilioretinal artery (other branching vessels that supply nutrients to other parts of the eye are not discussed here) (Karti et al., 2024). Next, the central retinal artery subsequently branches into subbranches that feed the inner retina (Karti et al., 2024). In the general population, the average thicknesses of arterioles and venules at this location are 152 and 233 μm , respectively (Yusufu et al., 2025a). There is variation in the exact anatomy, where most commonly there are four subbranches, less common variations are three or five or more subbranches (Tuma et al., 2025). The short posterior ciliary artery branches into the circle of Zinn-Haller and into multiple ciliary retinal arteries (Karti et al., 2024), which commonly pass through of the peripapillary retinal nerve fiber layer on the temporal side of the optic disc and supply blood to the

papillomacular region (Tuma et al., 2025; Schneider et al., 2021). There is also considerable variation in the anatomy of ciliary retinal arteries, which are more commonly observed in one eye (70-90%) than in both eyes (20-30%) (Schneider et al., 2021). Next, the subbranches of the central retinal artery and the cilioretinal arteries then branch into smaller vessels (terminal arterioles (sized $\sim 20\text{-}30\ \mu\text{m}$), precapillary arterioles (sized $10\text{-}20\ \mu\text{m}$), and capillaries ($\sim 9\ \mu\text{m}$ and smaller)) (Stehouwer, 2018). The capillaries then drain into postcapillary venules, which subsequently drain into the central retinal vein, which leaves the retina parallel to the artery through the optic nerve head (Sasongko et al., 2010; Karti et al., 2024). An overview of the organization of the microvasculature at different levels is shown in Fig. 1.

3.3. Retinal vessel wall composition

Before the central retinal artery leaves the optic nerve head to enter the retina, the vessel wall of the central retinal artery consists of several layers. From outermost to innermost layers, these are: tunica adventitia (which consists of collagen and elastic fibers), smooth muscle cells ("muscularis"; multiple layers), inner elastic lamina (which consists of elastic fibers), basement membrane, endothelial cells (one layer), and glycocalyx (which forms the barrier between endothelial cells and blood cells) (Sasongko et al., 2010). When the central retinal artery leaves the optic nerve head, the composition of the vessel wall changes. From here, retinal arterioles are covered by glia (astrocyte endfeet, Müller glial cell processes), the muscularis consists of only a single layer of smooth

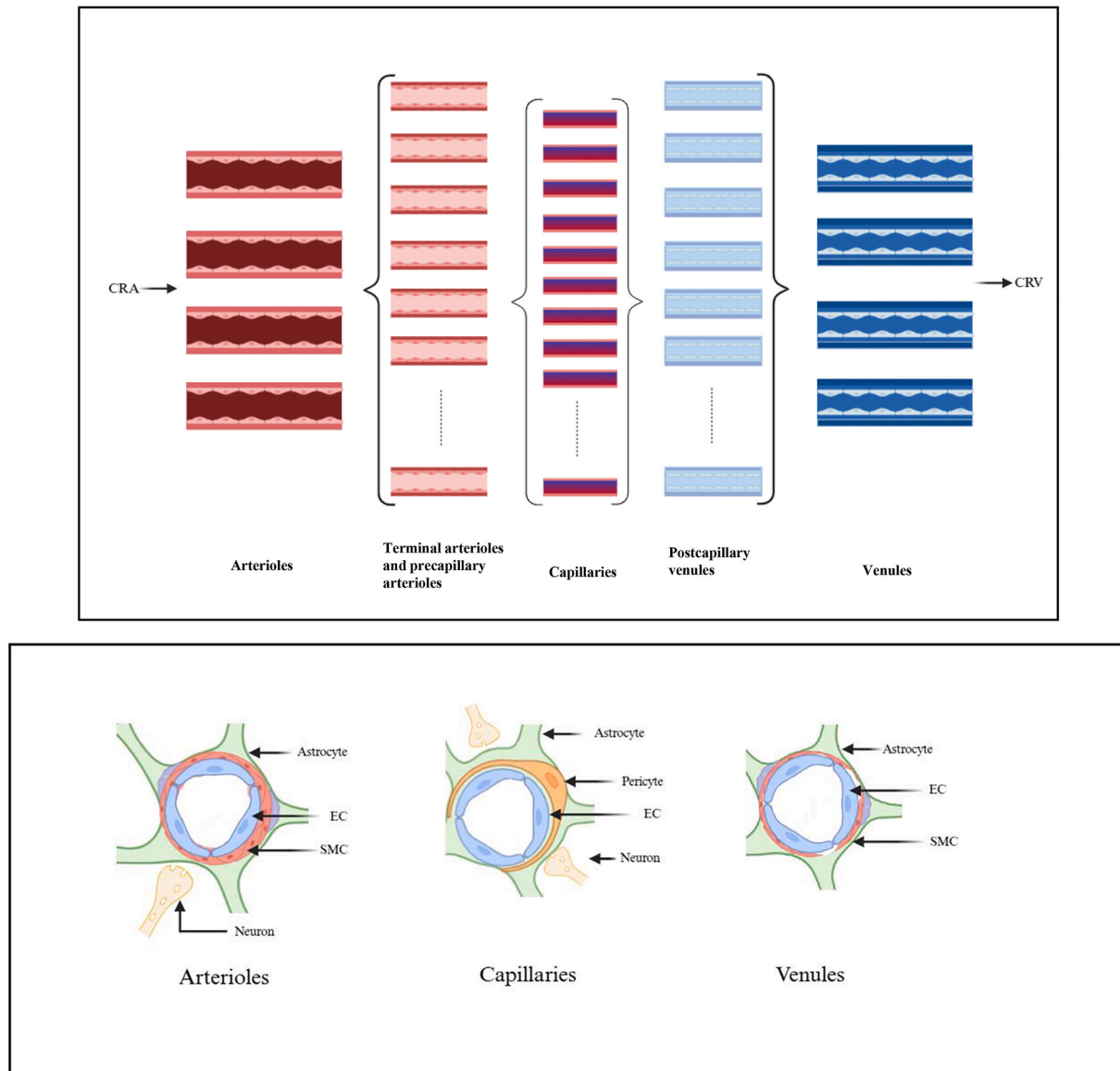


Fig. 1. Anatomy of the retinal microvasculature

Fig. 1.1 shows the organization of the retinal microvasculature. The central retinal artery branches into subbranches of arterioles that subbranch into terminal arterioles ($\sim 20\text{-}30\ \mu\text{m}$), precapillary arterioles ($\sim 10\text{-}20\ \mu\text{m}$), and capillaries ($\sim 9\ \mu\text{m}$ and smaller). Capillaries collect into many postcapillary venules ($\sim 10\text{-}30\ \mu\text{m}$), venules and the central retinal vein. Fig. 1.2 shows that the vessel wall differs at different levels of the microvasculature. Notable differences are the smaller amount of muscular smooth muscle cells in the vessel walls of venules than in the vessel walls of arterioles. In Fig. 1.2 pericytes are only shown at the level of the capillaries, but they are also found at other levels of the microvasculature. Pericytes are sparse at the level the arterioles, some pericyte-like mural cells are found in the pre-capillary arterioles, and high levels of pericytes are found at the levels of the capillaries. Not shown are other routes for supply of blood (i.e. retinal ciliary arteries) and removal of waste (glymphatic-like system and Müller cells) and the glycocalyx. Abbreviations: EC, endothelial cell; SMC, smooth muscle cell; CRA, central retinal artery; CRV, central retinal vein.

muscle, and the inner elastic lamina is significantly thinner or non-existent (Sasongko et al., 2010). At the pre-capillary arteriolar and capillary level, the arteriolar wall is very thin. It is formed by endothelial cells, smooth muscle cells, and pericytes, which are surrounded by astrocytic end-feet and a thin basement membrane (Fig. 1). (van der Heide et al., 2022) There is a gradual transition in the composition of the vessel wall from precapillary arterioles to arterioles, with smooth muscle cells becoming thinner towards the capillaries; and pericytes becoming more abundant (at the capillary level the ratio of endothelial to pericyte cells is 1:1 to 3:1) (D'Esposito et al., 2025). The blood-retinal barrier is located across precapillary arterioles, capillaries, and postcapillary venules (van der Heide et al., 2022; D'Esposito et al., 2025; Santisteban and Iadecola, 2025).

When capillaries coalesce to form postcapillary venules, the composition of the external wall changes; the vessel wall consists (from outermost to innermost layer) of thin fibrocytes and collagen fibers ("adventitial layer"), smooth muscle cells (sparser than in the arteriolar vessel wall), elastic fibers ("media"), endothelial cells, and glycocalyx (Sasongko et al., 2010; Kaur et al., 2022). The media in the postcapillary venules are very thin, and elastic fibers are sparse. The number of smooth muscle cells is relatively small, and these cells are not continuously present. Further, there are relatively few adventitial fibrocytes, and the collagen is thin.

3.3.1. Physiology

The microvasculature, sized between ~ 30 and ~ 300 μm , has two main functions: 1) to optimize the delivery of nutrients and removal of waste products in response to variations in demand (metabolic function) and 2) to decrease and stabilize pulsatile hydrostatic pressure at the level of capillaries (hemodynamic function) (Stehouwer, 2018). The hemodynamic function is thought to be mediated mainly by resistance arterioles and to a lesser extent by smaller vessels; the exact contribution of different levels of the vasculature to flow resistance is an active topic of research (Anderle et al., 2025).

3.4. Neurovascular coupling unit

Hemodynamic and metabolic functions are regulated by autor-regulatory mechanisms that involve the neurovascular coupling unit (Jobling et al., 2025). Fluctuations in pressure are sensed by vascular smooth muscle cells, which adjust their tension to regulate vascular resistance (Jobling et al., 2025). Regulation of blood flow to meet tissue metabolic demand is achieved through neurovascular coupling (metabolic autoregulation) (Jobling et al., 2025). Within the neurovascular coupling unit, bidirectional interactions occur between neural cells (neurons, astrocytes, microglia, oligodendrocytes) and cells that constitute the microvascular vessel wall, i.e., endothelial cells, pericytes, and vascular smooth muscle cells (van der Heide et al., 2022; Santisteban and Iadecola, 2025). In response to increases in metabolic demand of retinal neural cells, the inflow of blood from arterioles is increased via vascular and neurovascular mechanisms. One, neural cells that surround arterioles release nitric oxide and other compounds to induce relaxation of vascular smooth muscle cells. Two, neural cells can transmit signals to endothelial capillary cells, which are then sent to upstream arterioles via endothelial cell-to-cell connections to reduce vascular smooth muscle tension (via release of nitric oxide (NO) and other mediators), which increases blood flow (Arciero et al., 2013; Pappelis et al., 2020; Drachmann et al., 2025; Hein et al., 2012). Both mechanisms may be induced by increases in neural activity and changes in blood levels of specific molecules (e.g., higher carbon dioxide or other metabolites and lower oxygen levels). Pericytes may also contribute to the regulation of vascular smooth muscle cell tension by inducing NO release from neural cells (*vasculo-neuronal coupling*; e.g., via pericyte-astrocyte interactions) (D'Esposito et al., 2025; Santisteban and Iadecola, 2025). Furthermore, bidirectional interactions between neuronal and microvascular structures might occur at the level of venules (Santisteban and Iadecola,

2025).

3.5. Glycocalyx

The glycocalyx, which lines the luminal side of the endothelium, plays an important role in regulating endothelial cell functions, including mechanotransduction, leukocyte adhesion, and microvascular permeability (Inoda et al., 2025). Deterioration of the endothelial glycocalyx impairs microvascular function (Inoda et al., 2025).

3.6. Glymphatic-like system

Recent tracer studies in post-mortem eyes and animals indicate that clearance of waste products from the retina also occurs via a glymphatic-like system via anterograde and retrograde transport (Delle et al., 2024; Sun et al., 2025). Waste products that originate from neural tissue in the inner retina are thought to pass through peri-arterial and venular spaces and interstitial routes to the dural lymphatic vessels and cervical lymphatic routes through which fluid can (ultimately) be transmitted to the systemic circulation (Delle et al., 2024; Sun et al., 2025). The quantitative contribution of the glymphatic system to the overall clearance of waste products from the retina remains incompletely understood (Delle et al., 2024; Sun et al., 2025).

3.6.1. Differences between the central and peripheral retinal microvasculature

In this manuscript, we refer to the central retinal area as the posterior pole (visible on a 50-degree fovea-centered photo); and the adjacent part of the near and mid periphery, that is visible on a 50-degree optic-disc-centered photo (Fig. 2). (Quinn et al., 2019) We refer to the peripheral retina as the part of the retina outside this central retinal area (i.e., the near, mid, and far periphery) (Quinn et al., 2019).

3.7. Anatomy

The neurovascular organization of the central and peripheral retina differs as these regions serve different visual functions (Quinn et al., 2019). The central retina enables high spatial and photopic vision, while the retinal periphery enables low-resolution spatial vision, scotopic vision (Quinn et al., 2019). Towards the periphery, the retina becomes thinner. It contains a lower number of neurons, e.g., the density of ganglion cells decreases from 32,000–38,000 cells/ mm^2 in the central retina to 200–300 cells/ mm^2 at the ora serrata (i.e., the peripheral edge of the retina) (Quinn et al., 2019). This sparser neural organization has a lower metabolic demand and hence a less dense organization of the retinal microvasculature (Quinn et al., 2019).

3.8. (Patho)physiology

It has been suggested that there are differences in autoregulation between the central and peripheral retinal microvasculature (Bek and Jeppesen, 2021). For example, a physical activity-induced increase in mean arterial blood pressure was associated with a higher saturation of retinal venular drainage in the periphery but not in the central retina (Bek and Jeppesen, 2021). This may be due to shunting to circumvent the capillary beds ("steal phenomenon") (Bek and Jeppesen, 2021). Beyond autoregulation, the central and peripheral retina exhibit distinct physiological characteristics. The central retina (macular region) demonstrates approximately 50% higher blood flow than the peripheral retina, attributable to larger vessel diameters ($\sim 20\%$ greater) and faster red-cell velocity ($\sim 25\%$ faster). Vascular conductance is also $\sim 53\%$ higher centrally, reflecting its greater metabolic demand (Rassam et al., 1996). In contrast, the peripheral retina operates with lower flow and thinner vascular architecture (Rassam et al., 1996). Capillary organization further differs, the central retina contains multiple plexuses—radial peripapillary, superficial, intermediate, and

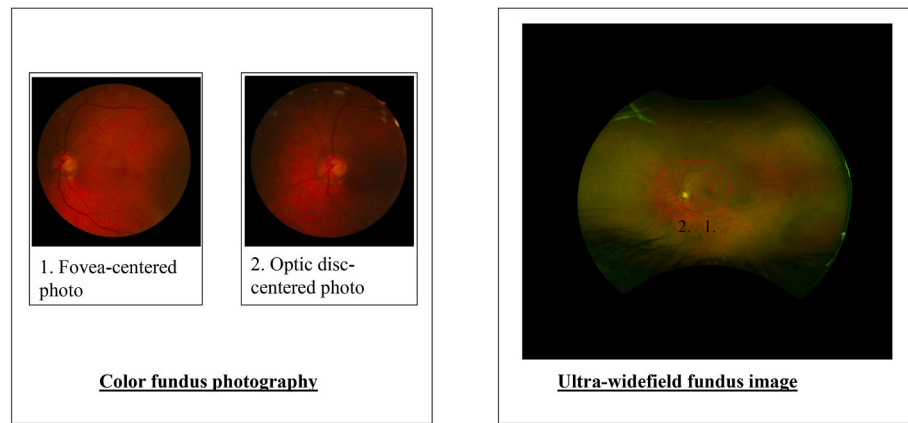


Fig. 2. Example images of color fundus photo and an (ultra)widefield fundus image

Fig. 2 shows fovea- and optic-disc centered images acquired with color fundus photography (AFC-230, Nidek Co, Gamagori, Japan) and an (ultra)widefield image (California, Optos plc, Dunfermline, Scotland) acquired on the same eye. The area of the retina that can be visualized with the (ultra)widefield camera is ≥ 6 times the size of the retinal area that can be observed with a 45-degree color fundus photo. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

deep—supporting fine perfusion control for the macula's high oxygen consumption (Lavia et al., 2020). These layers thin toward the periphery, where capillary density is markedly reduced (Lavia et al., 2020). Autoregulatory responses to metabolic changes, such as hyperoxia, are robust in both regions, but the central retina's higher baseline flow underscores its greater metabolic load (Rassam et al., 1996).

In addition, in the peripheral and central retina, different pathological vascular manifestations have been observed in ocular diseases such as diabetic retinopathy (Bek, 2013). In the central retina, impaired autoregulation is associated with the development of capillary hypertension and subsequent capillary leakage (resulting in microbleeds and exudates) (Bek, 2013). In the peripheral retina capillary occlusion is more frequently observed than in the central retina (Bek, 2013).

4. Section 2: Current methods

4.1. Image acquisition

4.1.1. Central retina

In many epidemiological studies, the central retina is imaged with color fundus photography centered on the optic disc or the fovea (Fig. 2). These are the standard images one and two from the seven-field stereoscopic fundus photography system (a standard method initially developed for diabetic retinopathy severity grading) (Govinda and de Verteuil, 2011). Most color fundus cameras enable a field of view between 25 and 60°. Dilatory eye drops (mydriasis) are commonly used both in epidemiological and clinical studies to allow visualization of a larger part of the retina to acquire images with better quality.

4.1.2. Peripheral retina

4.1.2.1. (Ultra)widefield cameras. The peripheral retina (up to a 200-degree field of view) is most commonly visualized in studies of peripheral retinal microvasculature using a scanning laser ophthalmoscope (SLO) from Optos (Optos plc, Dunfermline, Scotland) (Ashraf et al., 2021; Csincsik et al., 2018; Leung et al., 2024; Pellegrini et al., 2014, 2018). These instruments make use of red, green, and/or blue laser light and an internal ellipsoid mirror to visualize the retinal periphery (Quinn et al., 2019). A limitation is the use of an ellipsoid mirror, which leads to image distortion in the far temporal and nasal periphery. This can make vessels appear relatively larger than they are in reality (Quinn et al., 2019). A second limitation is that the resolution for imaging the central retina is lower than for most color fundus cameras (more details in Table S1) (Quinn et al., 2019).

4.2. Image analysis

Below, we outline the key processing steps for quantifying retinal microvascular imaging features. These are: vessel segmentation, arteriole and venule classification, anatomic landmark detection, region of interest selection, feature measurement and interpretation, pixel to micrometer conversion, and image quality assessment (Huang et al., 2016). All of these factors can influence the calculation of quantitative retinal microvascular imaging features (Huang et al., 2016). We present a brief overview of the general processing steps commonly used for the calculation of quantitative retinal microvascular imaging features.

4.2.1. Image processing

Overall image processing consists of three main steps (Kipli et al., 2018) i.e. 1) image pre-processing (including image cropping) and contrast improvement (Kipli et al., 2018); 2) vessel segmentation and arteriole and venule classification (Kipli et al., 2018); and 3) region of interest selection and calculation of retinal microvascular metrics (Kipli et al., 2018).

4.2.1.1. Image pre-processing and contrast improvement. Many retinal vessel analysis tools convert color images into grayscale images (Kipli et al., 2018). This step allows to keep information on key features that are later used in image processing (e.g., edges, regions, junctions) but reduces the amount of information that needs to be processed (Kipli et al., 2018). Grayscale conversion can be done using a combination of green, blue, or red channels or by selecting a single channel from the color image (Biswas et al., 2022). The green channel is commonly used as it provides the best image for visualizing the retinal vessels due to the high contrast between them and the retinal tissue (Biswas et al., 2022). With red and blue channels, certain parts of the retina can be better visualized. Choroidal vessels and optic disc edges are more clearly visible in the red channel (Biswas et al., 2022). The blue channel allows better visualization of retinal nerve fiber layer defects, epiretinal membranes, and lens flares (Biswas et al., 2022). Next, the image contrast is adjusted to reduce variations in luminosity and irregularities in contrast across the image. This step may also be used to enhance the contrast of specific features, such as edges. Various methods can be used for this, commonly used methods include spatial domain filtering and image restoration (Kipli et al., 2018; Zhang et al., 2024a).

4.2.1.2. Segmentation. After these initial preparation steps, the retinal microvasculature is segmented as well as image borders and anatomic landmarks are identified (using the edges of anatomical structures on the

image). Multiple methods for this exist, including AI-based methods, parallel multiscale feature extraction, and region growing (Kipli et al., 2018). Some of these methods allow for the correction of the spherical shape of the eye (van den Berg et al., 2025). The final result is a vessel segmentation map showing retinal arterioles and venules in different colors. The exact type of segmentation mask generated differs between tools (Kipli et al., 2018). One frequently used approach is to create a vessel skeleton, a 1-pixel-wide line that spans the centers of all detected vessels (Vargas Quiros et al., 2025; Zhou et al., 2022). This can be created in several ways, including a connected-component search strategy with single pixel evaluation, graph-based approaches with spline modeling, or morphological thinning methods that iteratively peel away boundary pixels while preserving the topology of the vessel network (Vargas Quiros et al., 2025; Zhou et al., 2022).

4.2.1.3. Region of interest assessment. Quantitative metrics are calculated in a specified region of interest to ensure the same biological structures are visualized across eyes. The main anatomic landmark used to select the area of interest is the optic disc (Vargas Quiros et al., 2025; Zhou et al., 2022; Yuan et al., 2024; Cheung et al., 2010).

For optic disc-centered photos, several regions of interest are commonly used in the literature. These include zone A (a circular zone between 0 and 0.5 disc diameter around the optic disc), zone B (a circular zone between 0.5 and 1.0 disc diameter around the optic disc), and zone C (a circular zone between 0.5 and 2.0 disc diameter around the optic disc; Fig. 3) (Ikram et al., 2013). These zones are widely used, but there are still different definitions in use, for example a recent study defined zone C as the region between 1.00 and 2.5 disc diameters from the optic disc border (Leveque et al., 2023).

Differences in zones originate from the historical development of tools for the quantification of retinal microvascular imaging features. The first quantitative assessments of retinal vascular parameters originated from the Atherosclerosis Risk in Communities (ARIC) Study in 1999 (Hubbard et al., 1999). This study developed a semiautomated

measurement method and the ARIC grading grid to evaluate retinal vessel caliber on 45° digitized fundus photographs (Hubbard et al., 1999). The grid consisted of two concentric zones: zone A and zone B, but vascular caliber measurements were derived exclusively from zone B (using a high-resolution display and early-generation image-analysis software).

Building upon the ARIC framework, the Singapore Malay Eye Study (SiMES) further refined and expanded retinal vascular quantification using a semiautomated, computer-assisted analysis program. This approach extended the original ARIC grid to include an additional region i.e., zone C. The broader analytical framework enabled the extraction of more complex vascular biomarkers, including branching patterns, fractal dimension, and vascular tortuosity. The software developed for this purpose at the National University of Singapore—later named the Singapore I Vessel Assessment (SIVA) system—integrated automated optic disc detection with user confirmation, automated classification and tracing of arterioles and venules, and algorithms capable of deriving a range of quantitative vascular metrics such as vessel diameters (or calibers), branching characteristics, tortuosity, and fractal dimension (Cheung et al., 2011).

The regions of interest used to calculate quantitative retinal microvascular metrics from fovea-centered photos and (ultra)widefield images are less well standardized. Some analysis tools analyze the vessel throughout the entire image, while others use only parts of the image. More details are provided in the section on analysis tool comparisons below.

The exact comparability of regions of interest between eyes is dependent on the field of view visible and the accurate detection of the optic disc for selecting the area of interest. The field of view depends on the extent to which image magnification occurs (which is determined by eye anatomy and camera-related factors) (Pors et al., 2024). Differences in the magnification factor between eyes can have a large impact on the exact field of view on the retina. For example, on a 45-degree fundus photo, there is 31% difference in retinal area visible between an eye with a refractive error of minus three diopters versus an eye with plus three diopters (Pors et al., 2024). Eye anatomy-related factors are estimated to determine ~47% of the magnification factor (axial length, cornea thickness and curvature, lens curvature, and spherical equivalent) (Pors et al., 2024). Camera setup is estimated to account for ~45% of the magnification factor (Pors et al., 2024).

4.2.2. Quantification of retinal microvascular image features

There are four types of retinal microvascular measurements that are widely used throughout the literature: 1) width-related measures, 2) tortuosity measures, 3) branching angle and related measures, and 4) measures of vessel complexity and area filling (i.e., fractal dimension, alpha shapes, and area measures) (Ikram et al., 2013; Li et al., 2016a; Vargas Quiros et al., 2025; Zhou et al., 2022; Pead et al., 2021). Exemplary images for all types of measures are shown in Figs. 3 and 4.

4.2.3. Width-related measures

4.2.3.1. Quantitative retinal microvascular measures. Primary measures of width that are frequently used throughout the literature are central retinal arteriolar and venular equivalents (CRAE, CRVE) and their ratio (arteriole-to-venule ratio (AVR)) (Ikram et al., 2013). CRAE and CRVE are commonly calculated from the width of the six arterioles and venules using the revised formula by Knudtson (Fig. 3). (Knudtson et al., 2003) To calculate this, the width is assessed for the six largest arterioles and venules, and then pairs are made of the widest, intermediate, and narrowest vessels. Next, for each pair, a hypothetical paired vessel width is generated (formula: $\sqrt{d_1^2 + d_2^2}/2$), giving three pairs (“p1-p3”). These pairs are subsequently combined into CRAE or CRVE. First, the two largest pairs (“p1” and “p2”) are combined into a new pair (“p4”), and then this new pair (“p4”) is combined with the narrowest pair, “p3”,

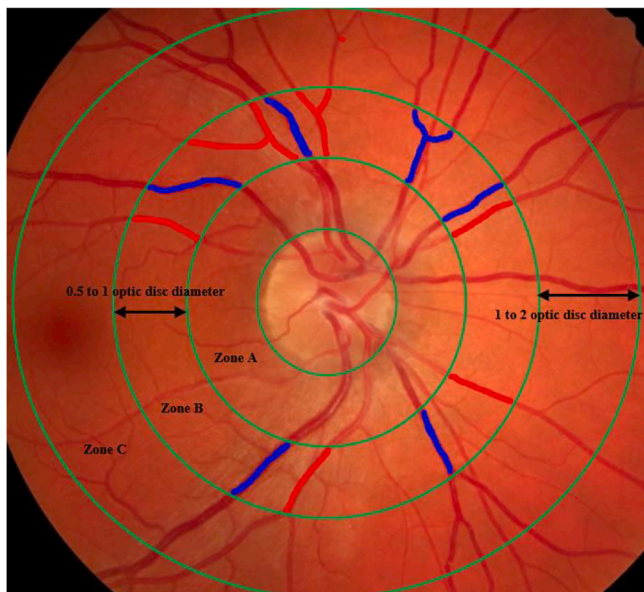


Fig. 3. Standardized regions of interest (zones A, B, and C) for optic disc-centered photos

Fig. 3 shows standardized zones A, B, and C which are commonly used as regions of interest when analyzing optic disc-centered photos (zone A: 0-0.5 disc diameter around the optic disc; zone B: 0.5-1.0 disc diameters around the optic disc; and zone C: 0.5-2.0 disc diameter around the optic disc). The six largest arterioles and venules in zone B are indicated in red and blue, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



directions (Ikram et al., 2013; Li et al., 2016a; Liew et al., 2008; Stehouwer, 2018). The interpretation of changes in retinal arteriolar and venular diameters is based on population-relative measures. There are no (widely accepted) absolute thresholds to determine ‘normal’ or ‘pathological’ retinal arteriolar or venular diameters, although some thresholds have been proposed for arteriole-to-venular diameter ratio (Hanssen et al., 2022).

4.2.4. Tortuosity

4.2.4.1. Quantitative retinal microvascular measures. The most widely used measures to quantify the curvature (“tortuosity”) of the retinal microvasculature are distance tortuosity (which is also referred to as simple tortuosity or arc-over-chord ratio), squared curvature tortuosity, and tortuosity density (Fig. 4). Distance tortuosity is quantified as the “real” length of the vessel between the start and endpoint divided by the shortest possible distance between the start and endpoint (the so-called Euclidean distance) using specified inflection points (Giesser et al., 2024). Squared curvature tortuosity quantifies the average curvature of a vessel (Giesser et al., 2024). Tortuosity density measures how much a vessel twists by quantifying curvature between curving points (so-called inflection points). The distance tortuosity is determined between inflection points, and overall tortuosity density is calculated by weighting distance tortuosity segments according to the proportion of the length of the entire vessel (Zhou et al., 2022). This measure is considered more accurate than distance tortuosity (Zhou et al., 2022).

Other tortuosity measures have also been proposed, but are less used throughout the literature (Chen et al., 2020; Krestanova and Penhaker, 2020; Lisowska et al., 2014). A few examples are inverse spherical radius tortuosity, angle-based tortuosity, and vasculature curvature index (Giesser et al., 2024). Inverse spherical radius tortuosity is calculated by averaging the radius of curvature at each point along the vessel and inverting this average value (Giesser et al., 2024). Angle-based tortuosity is calculated as the mean angle of deviations along the trajectory of the blood vessels (Giesser et al., 2024). Vascular curvature index is calculated by estimating angular momentum (Giesser et al., 2024). Differences in angular momentum between vessels with higher and lower tortuosity levels are calculated by estimating trajectories of curvature from an initial moment of inertia. This measure has been designed to maximize resistance against errors in vessel segmentation (Giesser et al., 2024). Fig. S1 shows image examples that depict the calculation of these measures.

4.2.4.2. Biological interpretation. Higher levels of tortuosity are thought to reflect higher levels of microvascular dysfunction. Degradation of retinal vessel walls (e.g., loss of elastin) or supporting tissue (e.g., collagen) is thought to predispose to the development of tortuosity, and the development of tortuosity can further be aggravated under hypertensive circumstances (Han, 2012; Weiss et al., 2020). Alternatively, tortuosity may also be induced by reductions in blood flow, which lower axial tension on the vessel wall (Han, 2012; Weiss et al., 2020).

4.2.5. Branching angle and related measures

4.2.5.1. Quantitative retinal microvascular measures. There are seven main measures: branching angle, branching coefficient, asymmetry ratio (or asymmetry factor), angular asymmetry, junctional exponent deviation, optimality, and length-to-diameter ratio (Ikram et al., 2013; Zhu et al., 2025b). Fig. 4.2 shows exemplary images that illustrate which parts of the retinal microvasculature are used to calculate branching angles and related measures.

A branching angle is formed by the two daughter vessels of a primary vessel (Fig. 4.2). The branching angle calculates the angle between two daughter vessels of a primary vessel that branch off ($\theta_1 + \theta_2$). The branching coefficient is calculated as the ratio of the branching vessel

widths to the primary (trunk) vessel width $(d_1 + d_2)^2/d_0^2$. Asymmetry ratio is defined as the ratio of the two branching vessel widths d_2^2/d_1^2 . Angular asymmetry is defined as the difference in angle between two branching daughter angles ($\theta_1 - \theta_2$). Junctional exponent deviation reflects the difference in the ratio of diameters at retinal vessel branch points in comparison with optimal organization, as defined from a healthy vascular network (the formula is $(|d_0^3 - d_1^3 - d_2^3|^{1/3})/d_0$). Optimality is calculated from symmetry between daughter and primary trunk vessel (the formula is $^3\sqrt{(d_1^3 - d_2^3)/(2 \times d_0^3)}$). Length-to-diameter ratio is calculated as the ratio of the length between two branching points to the trunk vessel width (L/d_0). The length is calculated between a first branching point and the closest second branching point of one of the daughter vessels from the first vessel branching point (Zhu et al., 2025b). Other measures for quantification of retinal microvascular angles have also been proposed, but are less widely used (Wang et al., 2024).

4.2.5.2. Biological interpretation. Differences in branching angle and related measures are thought to reflect deviations from optimal branching organization, which is considered to reflect suboptimal organization of blood flow and higher levels of microvascular dysfunction (Ikram et al., 2013).

4.2.6. Measures of vessel complexity and area filling

4.2.6.1. Quantitative retinal microvascular measures. We outline three types of measures that quantify the complexity of vessel organization and/or area filling (i.e., fractal dimension measures (Fhima et al., 2023), alpha shapes (Pea et al., 2021), and area measures) (Vargas Quiros et al., 2025; Zhou et al., 2022).

Multiple fractal dimension measures exist, including D_0 (capacity), D_1 (entropy), D_2 (correlation), singularity length, asymmetry of the $f(\alpha)$ - α spectrum, and lacunarity (Fhima et al., 2023). D_0 is a measure of the extent to which an area is filled with vessels; D_1 is a measure of overall spatial complexity across the image; D_2 is a measure of vessel concentration (i.e. vessel grouping) throughout the image; singularity length is a measure of the range of the complexity of the vessel organization; asymmetry of the $f(\alpha)$ - α spectrum is an index of the shape of the spectrum of complexity; and lacunarity is a measure of the homogenous distribution of the vessels (Fhima et al., 2023). The measures are calculated using a box-counting method, which is explained in the legend of Fig. 4.3 (Fhima et al., 2023).

Other measures that estimate complexity or image filling are alpha shapes, vessel area, and vessel density (Vargas Quiros et al., 2025; Zhou et al., 2022; Fhima et al., 2023; Pea et al., 2021). Alpha shapes are used to calculate vessel complexity ($Op_{\alpha_{min}}$), vessel spread (OpA), and vessel shape (VS) (Pea et al., 2021). $Op_{\alpha_{min}}$ is calculated by determining the minimal size of a so-called alpha shape (i.e., Delaunay simplices (e.g., triangles)) that is required to cover the entire vessel segmentation map (Pea et al., 2021). OpA is calculated as the total area spanned by the vascular network (Pea et al., 2021). VS is calculated as the ratio between $Op_{\alpha_{min}}$ and OpA (Pea et al., 2021). Vessel area is calculated as the amount of two-dimensional area filled by vessels. Vessel density is calculated as the vessel area divided by the total area visible (Vargas Quiros et al., 2025; Zhou et al., 2022; Fhima et al., 2023).

4.2.6.2. Biological interpretation. Sparser retinal microvascular networks are thought to reflect higher levels of microvascular deterioration and can be assessed as lower complexity, higher levels of asymmetry (fractal dimension measures) (Ikram et al., 2013), and lower $Op_{\alpha_{min}}$, OpA , or VS. (Pea et al., 2021)

4.2.7. Pixel to micrometer conversion

Width-related measures are calculated in pixels from the images. A conversion is needed to express the measurements μm (Schanner et al.,

2023). The conversion factor is different for the central and peripheral retina because the retina is a convex surface, and this difference becomes larger as the field of view increases (Pellegrini et al., 2014).

For the central retina, two methods are regularly used for the conversion of vessel width from pixels to micrometers using the optic disc size as a reference point (Schanner et al., 2023). One method uses a group average conversion factor ("generalized approach"), and one method uses a conversion factor that differs per eye ("individualized method"). In the generalized approach, the conversion factor is determined by dividing 1800 or 1850 μm by the average optic disc size of a certain group of individuals. The average optic disc diameter for an adult with an emmetropic eye and axial length of 24 mm is 1800 or 1850 μm (Schanner et al., 2023). In the individualized approach, the conversion factor is calculated for each eye separately. The standard disc diameter (1800 or 1850 μm) is divided by the optic disc size measured in pixels for that eye. Both approaches have been compared in a (small) recent study, which found that the individualized approach may be more accurate.¹⁰⁰ Additionally, the conversion factor for pixel to μm conversion can also be determined using the distance between the disc and the fovea (i.e., the disc-fovea distance) (Kwak et al., 2025; Chiang et al., 2021). This method has been proposed to assess the degree of retinal vascularization for the assessment of disease severity in retinopathy of prematurity (Kwak et al., 2025; Chiang et al., 2021).

For the peripheral retina, the pixel-to- μm conversion factor differs according to the exact location of the retina where the microvasculature is detected. For example, one pixel has been found to correspond to 17.2 μm in zone B and to 20.6 μm in the retinal periphery (4 optic disc diameters away from the optic disc) (Pellegrini et al., 2014).

It is important to realize that no direct method is known to measure the conversion factor. All the methods above should therefore be regarded as working approximations. The numerical values obtained by converting pixels to μm carry an unknown uncertainty.

4.2.8. Image quality

Several types of errors can negatively impact the visibility of the retinal microvasculature (Li and Chen, 2023). Common errors are the presence of artefacts (e.g. eye lashes, lid, or cataract), illumination errors (e.g. small pupil size and/or camera misalignment, reflection artefact), blurriness of the image (e.g. due to media opacities such as cataract, floaters), and wrong field of view (e.g. due to incorrect fixation on the visual target in the fundus camera) (Li and Chen, 2023). Examples of each type or error are shown in Fig. S2. The most common type of error on fundus images is a dark image due to small pupil size and media opacity issues (usually cataract). On (ultra)widefield images a common issue is the presence of eyelashes/eyelids and reflection artefacts. SLO imaging is less prone to artefacts caused by cataract.

Several clinical grading systems have been proposed for assessing image quality in studies of quantitative retinal microvascular imaging features. A commonly used classification is the eye-quality image quality grading system, which was originally developed for diabetic retinopathy screening (Shi et al., 2022; Vargas Quiros et al., 2025; Zhou et al., 2022). This classification system has six image quality categories (i.e., excellent, good, adequate, insufficient for full interpretation, insufficient for any interpretation, other). Image quality assessment criteria are illumination, image field definition, and the presence of artefacts. Excellent is defined as the absence of any errors that affect image quality. Good is defined as some problems in one or two image quality factors. Adequate is defined as some problems in three or four of the image quality factors. Insufficient for full or any interpretation indicates that the image could, respectively, not be assessed at all or only partially. Other clinical image grading systems have also been developed and used in studies that investigated quantitative retinal microvascular imaging features (Lamirel et al., 2012).

A second image quality assessment system is the FundusQ-Net score. This is a 10-point image classification scale (range 1-10 points) developed for grading of the optic disc and the peripapillary region

(Abramovich et al., 2023). A high grade is given if the details of this area were clearly visible, and the score was decreased if the visibility was compromised (either due to resolution, focus, contrast, brightness, artefacts, overall haze, uneven illumination, or ability to detect fine details). However, this score does not provide insight into the reason for the reduced visibility of the retinal microvasculature.

Other image quality classification methods have also been proposed (e.g., for use in AI-based pipelines for disease grading). However, in this review, we do not outline all existing image grading systems as our focus is on image quality classification systems used in combination with tools that quantify retinal microvascular imaging features.

4.3. Comparison of analysis tools

We conducted a literature review to identify commonly used retinal microvascular analysis tools and compare them.

4.3.1. Scope

Our objective was to provide insight into heterogeneity in validation and reporting of tools that are available to (clinical) researchers for quantification of the retinal microvasculature. We focused on commonly used tools (i.e., software programs that consisted of pipelines that enable all processing steps from vessel segmentation to retinal microvascular measure calculation). Given the scope of this review, we focus on commonly used tools and do not review tools or algorithms that were only evaluated for the execution of one of the processes involved in retinal microvascular measure calculation (such as vessel segmentation) (Mahapatra et al., 2024). Also, we do not review microvascular imaging features specific to relatively rare diseases in the general population (such as those of hypertensive retinopathy) (Cheung et al., 2022a).

4.3.2. Literature search

We identified tools from the literature by searching PubMed up to November 2025. We searched for studies that provided insight into the development and validation of retinal microvascular analysis tools (including studies that compared different analysis tools). Keywords used to identify studies were related to retinal microvascular measures (width-related measures, tortuosity measures, branching angle, fractal dimension, alpha shapes, and vessel area) and clinometric properties (validation, test-retest, vessel annotation, Dice score). Commonly-used tools were also identified by extensively evaluating studies that reported on retinal microvascular measures in the context of ocular and systemic disease (including systemic risk factors). Key words used were: diabetic retinopathy, age-related macular degeneration, glaucoma, cardiovascular disease (stroke, heart failure, atrial fibrillation, carotid artery stenosis, coronary artery disease, coronary heart disease, peripheral artery disease), chronic kidney disease, Alzheimer's disease and related dementias, cognitive impairment, depression, Parkinson's disease, schizophrenia, mortality, cancer, chronic obstructive pulmonary disease, asthma, sociodemographic (age, sex, education level, occupation level), genetics, physical activity, sleep, smoking, hypertension, diabetes, obesity, body-mass index, cholesterol, stress, air pollution, arterial stiffness, sleep apnea, diet, inflammation). We compared the reporting and validation of analysis tools for vessel segmentation, arteriole and venule classification, detection of optic disc as an anatomic landmark, selection of the region of interest, selection and calculation of quantitative retinal microvascular measures, conversion of pixels to micrometers, and image quality assessment. As such, this review provides an overview of a large number of commonly used microvascular analyses tools that are currently available to clinical researchers.

4.4. Results

Table 1 shows an overview of the fifteen analysis tools that are compared in this review. There were eight semi-automatic methods and seven fully-automatic methods included. There were two vessel analysis

Table 1
Overview of commonly used tools for quantification of retinal microvascular imaging features.

	Year	Field of view	Validation of vessel segmentation (measures)	Validation OD position	Region of interest for OD-centered image	Region of interest for fovea-centered image	Measures	Image quality development purpose	Open source	Reference
Semi-automatic tools										
SIVA	2010	C, W	OD (ICC)	NA	C: Zones B,C W: Zone C, 0.5-7.0 DD	A circle with 5 times the optic disc diameter, (medial border at 1/4 of a disc diameter nasally from the nasal optic disc border	Wi, FD, To, Br	Retinal vessel analyses	No	C (Cheung et al., 2011) (Thomas et al., 2014), W (Leung et al., 2024)
IVAN	2004	C	OD (ICC)	NA	Zones B,C	-	Wi	Retinal vessel analyses	No	Mautuit et al. (2022)
VAMPIRE	2011	C	F (Dice)	NA	Zones B,C, or main vessel per quadrant	Main vessel per quadrant	Wi, FD, To, Br	DR screening	No	(De Abreu et al., 2025; Perez-Rovira et al., 2011)
VAMPIRE	2014	W	W (predictive value, delta)	NA	(ultra)widefield: 1) Annular segment between 3.25-4.25-disc diameter nasal to optic disc; 2) Two regions of interest: 50° (circle) and 120 × 85° (ellipsoid shape)		Wi, FD, To,	Retinal vessel analyses	No	(Csincsik et al., 2018; Pead et al., 2021, 2023; Robertson et al., 2020)
MONA REVA	-	C	OD (delta)	NA	Zone C	A circular shape without the center was positioned on the central fovea (0.75-2.5disc diameter in size)	Wi, FD, To	Clinical setting (emergency room), retinal vessel analysis	No	(Salon et al., 2023), (Van Craenendonck et al., 2021; Atiskova et al., 2021)
	-	C	OD (delta)	NA	Zone A	-	Wi	Clinical setting (emergency room)	No	French and Heitmar (2021)
ALTAIR	2020	C	OD (ICC)	NA	Zone B	-	Wi	-	No	Maderuelo-Fernandez et al. (2020)
Optos AV Annotator	-	W	W (correlation)	NA	(ultra)widefield: Inner zone: 1 to 1.5-disc diameters from the optic disc center Outer zone: 3.25 to 4.25-disc diameters from the optic disc center		Wi	Retinal vessel analyses	No	Ashraf et al. (2021)
Fully automatic tools										
SIVA-DLS	2021	C	OD (ICC)	-	Zones B,C	-	Wi, FD, To, Br	Retinal vessel analyses	No	Cheung et al. (2021)
QUARTZ	2015	C	F,OD (Se,Sp,Ac, predictive value)	Yes	Full image	Full image	Wi, area	Retinal vessel analyses	No	Fraz RAW et al. (2015)
RVPQMS	2025	C	F (Dice)	-	Zone B, Full image	Full image	Wi, FD, To, Br	Retinal vessel analyses	No	Zhu et al. (2025b)
RMHAS	2021	C	F,OD (Ac,Se, Sp, F1)	-	Zone B, Full image	Full image	Wi, FD, To, Br	Mix of DR screening and retinal vessel analyses	No	Shi et al. (2022)
AutoMorph	2022	C	F,OD (Se, Sp, prediction, Ac, AUC, F1)	Yes	Zone B,C, Full image	Full image	Wi, FD, To	DR screening	Yes	Zhou et al. (2022)
VascX	2025	C	F,OD (Dice)	Yes	Full image	Full image	Wi, Area, To, Br	DR screening	Yes	Vargas Quiros et al. (2025)
LUNet/PVBM	2024	C	OD (Dice)	-	Full image	Full image	FD, Br	Optic disc and peripapillary region visibility	Yes	(Fhima et al., 2023, 2024; Abramovich et al., 2023)

Table 1 shows commonly used tools for quantification of retinal microvascular imaging features. Two versions of VAMPIRE software were evaluated, one version was validated for analyses of the retinal microvasculature from color fundus photos; and one version for analyses of the microvasculature in the peripheral retina. Segmentation and arteriole and venule classification were expressed as Dice score (color fundus) or as true, false, positive and negative predictive values (and average differences ((ultra)widefield imaging)). Abbreviations: F, fovea; OD, optic disc; C, center; W, widefield; NA, not applicable; Wi, width measures; FD, fractal dimension or alpha shape; To, tortuosity; Br, branching angle and related measures; DR, diabetic retinopathy.

tools developed by the Vessel Assessment and Measurement Platform for Images of the Retina (VAMPIRE; one tool for analyses of color fundus photos and one tool for analyses of (ultra)widefield photos). We evaluated two SIVA analysis tools, one semi-automatic tool (referred to as “SIVA” below) and one fully automatic tool (“SIVA-DLS”). We considered LUNet and PVBm together as one tool as they have been developed by the same research groups and together provide a pipeline that covers all steps required for retinal microvascular imaging feature quantification. Among the fifteen tools, thirteen tools have been used on color fundus photos (fovea- and/or optic disc-centered photos); and three semi-automatic methods have been used on (ultra)widefield images (the semi-automatic SIVA tool was used on both color and (ultra)widefield images).

4.4.1. Validation of vessel segmentation and arteriole-venule classification

There was a considerable difference in the reporting and validation of the analysis tools (a detailed overview of validation results is provided in [Tables S2 and S3](#)). One, the retinal microvascular measures used to show the performance of tools for vessel segmentation and classification differ across studies. Five tools reported intraclass correlation coefficient or a different correlation measure, four tools reported Dice similarity coefficient (“Dice score”), for three tools the mean differences in width measures were reported, and for four tools relative performance measures were reported (i.e. sensitivity, specificity, positive predictive value, negative predictive value, accuracy, area under the curve and/or F1 score). Two, the fields of view tools were validated on different tools. Eleven tools were validated on optic disc-centered photos, six tools were validated on fovea-centered images, and two tools were validated on

(ultra)widefield images. Three, the ground truth used to compare the performance of tools differed. Eight tools were validated against manually annotated images (a comparison was made between the segmentation of the entire vessel segmentation map by the tool and expert manual annotation). For seven tools, insight into the performance of the tool was available only for a specific region of interest and only for certain measures (CRAE, CRVE, AVR). For these tools, we did not identify a comparison with a manually annotated image. One of the tools was used on (ultra)widefield and color fundus photos, but was only validated on color fundus photos (SIVA).

4.4.2. Anatomic landmark validation

Of the seven fully automatic analysis tools, three tools reported the accuracy of optic disc detection ([Table S4](#)). One study expressed performance as detection accuracy as success rate or percentage overlap with the optic disc (QUARTZ), one study reported Dice score (VascX), and one study reported sensitivity, specificity, positive predictive value, negative predictive value, accuracy, area under the curve, and F1 score (AutoMorph). In addition, one tool also reported accuracy for the detection of the fovea (VascX). Comparison of the performance of semi-automatic tools for optic disc detection is not meaningful, as these tools require manual input to detect optic disc position.

4.4.3. Region of interest

Six semi-automatic tools and four fully automatic tools analyzed quantitative retinal microvascular measures in zones B or C of optic disc-centered photos ([Tables S5–6](#)). Two semi-automatic tools (SIVA, MONA REVA) used a circular region of interest with a size of up to five optic

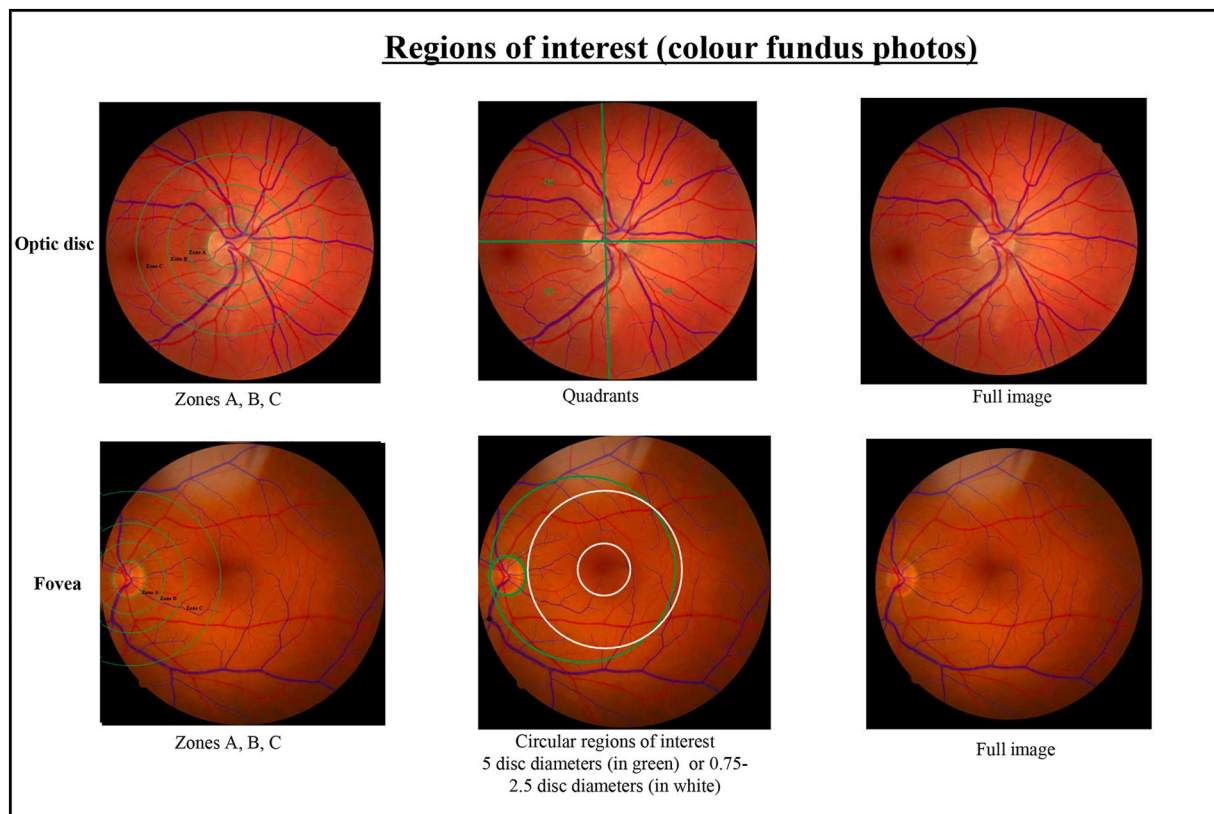


Fig. 5. Regions of interest used for color fundus photos

[Fig. 5](#) shows on the top row, from left to right, for optic disc-centered photos the following regions of interest: 1) zones A,B, and C, 2) quadrants 3) full image (the edges of the image determine the exact region of interest). On the bottom row, from left to right, for fovea-centered photos the following regions of interest are shown: 1) zones A, B and C (all of which are incompletely visible due to the edge of the image on the nasal side of the optic disc), 2) circular region of interest sized five optic disc diameters with border positioned 0.25 optic disc distance nasal from the optic disc (in green); or a circle without center of 0.75 to 2.5 disc diameters that is positioned on the fovea (in white); and 3) the full image (in which the edges of the image determine the field of view). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

disc diameters to analyze fovea-centered photos (Fig. 5). Six fully-automatic tools used the entire retinal image as a region of interest. One semi-automatic tool (VAMPIRE) analyzed one main retinal vessel per quadrant.

Three tools have been used to analyze the vasculature in the retinal periphery (SIVA, Optos AV annotator, and VAMPIRE) in varying regions of interest (Fig. 6). SIVA software has been used to study retinal vessels in an area between 0.5 and 7.0 disc diameters around the optic disc (Leung et al., 2024). The Optos AV annotator has been used to analyze the retinal microvasculature in a zone located 3.25 - 4.25 disc diameters from the nasal side of the optic disc (Ashraf et al., 2021). VAMPIRE software has been used to quantify retinal microvascular imaging features in: 1) a specified zone at 3.25- to 4.25-disc diameter, nasal to the optic disc (Robertson et al., 2020); 2) a circular area of 50° and an ellipsoid area of 120 × 85° (Pead et al., 2023; Joseph et al., 2024; Mitchell et al., 2025); and 3) each quadrant of the peripheral retina (where, in each quadrant, a main vessel was detected that was followed until the edge of the image) (Pead et al., 2023); and 4) midperiphery and beyond that, all visible area (Csincsik et al., 2018). In these studies, comparisons were made with regions. Additionally, the (ultra)widefield tool developed by VAMPIRE incorporated a method developed by Optos to enable accounting for the curvature of the retina.

4.4.4. Quantitative retinal microvascular measures

Quantitative retinal microvascular measures calculated by tools are outlined below. All tools calculated measures separately for arterioles and venules.

4.4.5. Width-related measures

All analysis tools calculated retinal microvascular width measures except PVBM (Table 1, S5, S6). The most commonly used measures were CRAE, CRVE, and/or AVR (these were used by ten tools, all of which

used the revised formula by Knudtson) (Knudtson et al., 2003). There were also ten tools that calculated the mean or median width over specified vessel segments. Smaller numbers of tools reported use of other width measures, including standard deviation of vessel width (five tools), minimal or maximal diameter (one tool, i.e. QUARTZ), or width gradient (two tools by VAMPIRE, i.e. both tools for analyses of color fundus and (ultra)widefield images).

4.4.6. Tortuosity

Mean or median distance tortuosity (i.e., simple tortuosity or arc-over-chord ratio) was calculated by eleven tools. Six tools calculated curvature tortuosity (or squared curvature tortuosity), and one analysis tool calculated tortuosity density (AutoMorph). The formulas used among open source tools (AutoMorph, PVBM) were highly similar, only for curvature tortuosity; the formulas used by VascX and PVBM were slightly different (VascX calculated curvature relative to the length of the vessel from which it was calculated; AutoMorph did not express results relative to the length of the vessel from which it was calculated).

4.4.7. Branching angles and related measures

Six tools calculated mean or median branching angles, and five tools calculated branching coefficient, asymmetry ratio, and/or junctional exponent deviation. Four tools calculated angular asymmetry. One tool calculated optimality (RVPQMS). The methods reported for the calculation of these measures were generally similar across studies. However, comparison of two open-source tools showed that branching angle results generated by VascX and PVBM were somewhat different (due to differences in the selection of angle and exact methods used to calculate angles; more details are given in Fig. S3).

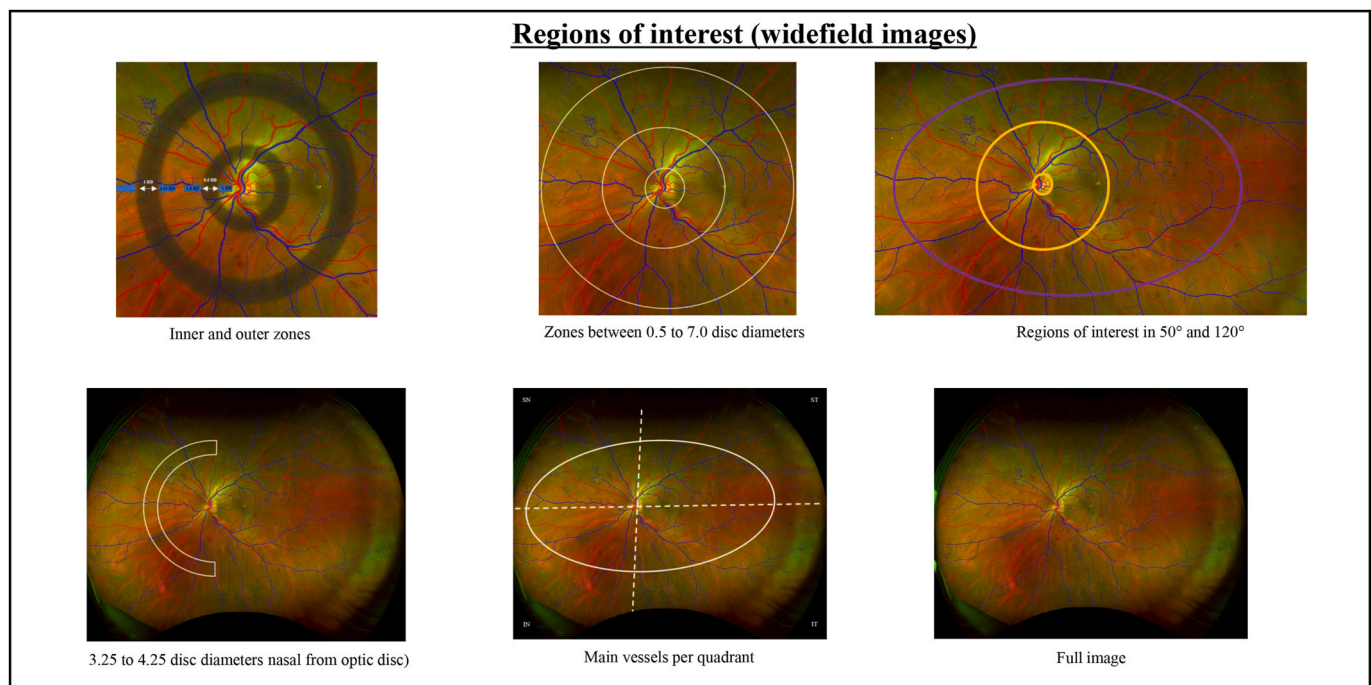


Fig. 6. Regions of interest used for (ultra)widefield fundus images

Fig. 6 shows from left to right on the top row: 1) inner and outer retinal zone's (defined as: 1 to 1.5-disc diameters from the optic disc center and 3.25 to 4.25-disc diameters around the optic disc center, respectively; in these regions the width of the three largest vessels was analyzed); 2) the zone between 0.5- and 7.0-disc diameters around the optic disc; and 3) zones of 50° (yellow) and 120 × 85° degrees (purple). From left to right on the bottom row are shown 1) a half-ring shaped region of interest between 3.25 and 4.25 disc diameters on the nasal side of the image; 2) a quadrant-based selection of region of interest (where selected vessels were followed as far as possible towards the edge of the image if possible); and 3) the full image is shown (in which the edges of the image and the presence of artefacts (e. g. lashes) determine the field of view). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

4.5. Fractal dimension, alpha shapes, and area measures

Ten tools calculated D_0 (Capacity dimension), three tools calculated D_1 (Entropy Dimension) or D_2 (Correlation Dimension). Singularity length was calculated by two tools, lacunarity was calculated by one tool (MONA REVA), and alpha shapes (Op α -min and Op α) were calculated by one tool (i.e. the VAMPIRE (ultra)widefield analysis tool). Vessel area covered by vessels or vessel density was reported by six tools. Comparison of the open source tools shows that the calculation of these measures was similar across tools.

Besides the main measures outlined above, certain additional variables are created, which serve to enable more insight into the exact calculation of quantitative retinal microvascular imaging features (e.g. the number of branching vessels). These additional variables are shown in Tables S5 and S6.

4.5.1. Pixel to micrometer conversion

Three semi-automatic tools (SIVA, IVAN, MONA REVA) reported conversion of pixels to micrometers in the central retina using an average disc diameter estimate of 1800 μm . One tool (VAMPIRE) determined a conversion factor using an individualized approach (optic disc size was assessed with optical coherence tomography) (De Abreu et al., 2025). None of the fully automatic tools reported the method for the conversion of pixels to micrometers.

Two of the three tools used for analyses of the peripheral retinal microvasculature reported the use of different conversion factors for the central and peripheral retina (i.e., VAMPIRE and Optos AV annotator).

4.5.2. Image quality

There was a considerable variation in the exact image quality criteria used across studies using microvascular analysis tools (details are provided in Table S7). Seven tools were used in combination with image criteria that were originally developed for clinical settings (either for use in an emergency room setting or for diabetic retinopathy screening) or for assessment of the optic disc and peripapillary region. Furthermore, the exact criteria used to determine image quality considerably varied. Some studies excluded images with only one type of error (e.g., the presence of artefacts), and some studies excluded images with one or more errors.

5. Section 3: Proposed framework for standardization

We propose a framework for standardizing the quantification of retinal microvascular imaging features. A summary of the framework is shown in Fig. 7 and in Table 2. We also provide recommendations for standardizing statistical analyses (Table 2).

5.1. Vessel segmentation and arteriole-venule classification

The performance of vessel segmentation and classification of arterioles and venules is dependent on the exact camera and field of view. Hence, ideally, the performance of vessel tools should be evaluated per camera separately and for fovea- and optic-disc-centered images separately (whilst using standardized criteria for assessment of image quality across studies (more details in the image quality section below)). Further, it is important to have insight into the performance of the algorithms across different ethnic populations and between eyes with and without retinal pathology. Indeed, quantitative retinal microvascular imaging features have been found to differ according to ethnicity (Wentzel et al., 2020; Rochtchina et al., 2008; Li et al., 2013) and between eyes with and without retinal pathologies (Eid et al., 2025). Hence, we recommend that tools are separately validated according to ethnicity and for eyes with and without ocular pathologies.

We further recommend that the performance of an algorithm for vessel segmentation and arteriole-venule classification be validated against a manually-drawn vessel segmentation map (annotated by or

Key steps for standardization of retinal microvascular imaging biomarker quantification

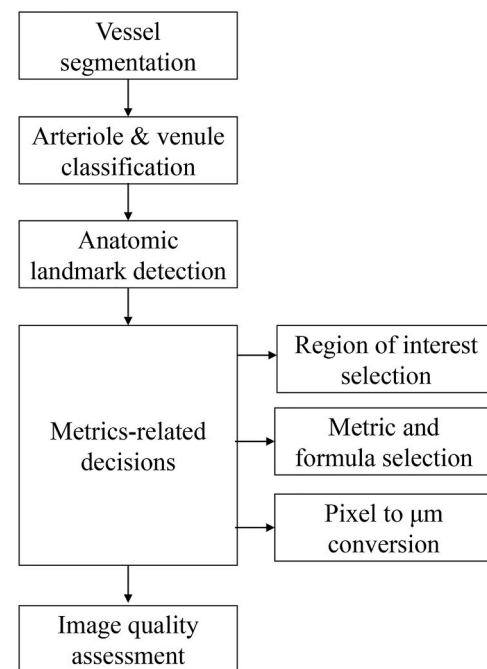


Fig. 7. Flowchart of key steps for standardization in retinal microvascular imaging feature quantification.

under the supervision of an experienced image grader or using a fluorescein angiography-derived image as ground truth). We recommend annotating the entire image (i.e., all vessels) and making several versions. For the first version, we recommend that retinal and arterioles be drawn in two separate colors (where vessels cross, the overlaying vessel should be drawn). For the second version of the vessel segmentation map, we recommend annotating in a third color where the retinal arterioles and venules cross (only crossing sections need to be annotated). This second version of the segmentation maps will enable to compare the performance of the tools for assessing vessel connectivity for vessels that may not appear connected, as they are overlaid by other vessels. In cases where arteriole and venule classification is difficult, researchers may also consider creating a third manual segmentation map in which vessels that are difficult to classify are indicated in a separate color (this enables evaluation of tools at different levels of certainty of vessel classification).

Interpretation of the performance of analysis tools is facilitated by the use of the same reporting quantitative retinal microvascular measures across studies. The reporting of Dice and vessel connectivity indices (such as centerline Dice scores) can be recommended as measures that enable objective comparability. Dice scores allow for the detection of errors in the classification of a vessel, arteriole, or venule at a pixel level. Centerline Dice scores can detect errors in connectivity detection within vessels (Eelbode et al., 2020). Alternative methods for quantifying connectivity have also been proposed, such as the Wasserstein distance (used to quantify optimal transport) (Kolouri et al., 2017). Performance indices that are less suitable for comparison are those indices that are sensitive to the relative distribution of errors in vessel segmentation or arteriole-venule classification across studies (e.g., sensitivity, specificity, accuracy, F1 score, area under the curve, positive predictive value, or negative predictive value) (Ahlbom, 2021). The relative distribution of errors may differ according to the camera or population used, or according to the number of images with poor image quality. Further, it is recommended to report the exact version number of the software used.

Although there are increasingly more open-source annotated

Table 2

Framework for standardization of quantification of retinal microvascular imaging features and conducting statistical analyses.

	Recommendation	Future research
1. Vessel segmentation	Report Dice score and vessel connectivity indices (centerline Dice score or Wasserstein distance) using manual annotation of the entire image as ground truth. Report software version number	Report Dice scores and vessel connectivity indices stratified per camera, population, and disease (e.g., DR, AMD, (pathological) myopia, glaucoma)
2. Arteriole/venule classification	<u>Minimal method:</u> Report Dice score and vessel connectivity indices (centerline Dice score or Wasserstein distance) using manual annotation of the entire image as ground truth. <u>Additional method:</u> Determine Dice score and vessel connectivity indices (centerline Dice score or Wasserstein distance) when omitting vessels for which there is doubt on arteriole and venule classification	Report Dice scores and vessel connectivity indices stratified per camera, population, and disease (DR, AMD, (pathological) myopia, glaucoma, other). Increase open source availability of annotated image datasets for validation to allow for validation of algorithms for different cameras, across different populations and different retinal diseases
3. Anatomic landmark detection	Report Dice score for optics disc detection using manually annotated images	Report Dice score per camera, population, and stratified by disease (DR, AMD, (pathological) myopia, glaucoma, other)
4. Measures-related decisions		
- Region of interest	Use of a standardized region of interest. For optic disc-centered photos, this could be zone C (0.5–2.0DD from optic disc). For other photo orientations (fovea-centered and (ultra)widefield images), the optimal region of interest needs to be defined.	Determine optimal region of interest from fovea-centered and (ultra) widefield images. -Consider use of magnification factor correction or multiple anatomical landmarks (e.g., fovea and optic disc) -Consider accounting for the curved shape of the retina
- Formula	Determine a minimal set of quantitative retinal microvascular measures and standardized formulas for use across tools. A minimal set could contain; -CRAE, CRVE, AVR (calculated using Knudtson's revised formula) -fractal dimension (D_0), simple tortuosity, mean branching angle	Compare differences between formulas across studies and consider the development of a composite score that combines multiple quantitative retinal microvascular measures. Determine interpretation and performance of metrics in the context of ocular and systemic disease. Development of novel quantitative retinal microvascular measures, e.g., using machine-learning-based methods
- Pixel to μm conversion factor	Use of different conversion factors according to the central or peripheral location of the retinal microvascular vessels assessed <u>Minimal method:</u> Use a 'generalized' (i.e., non-individual) method for pixel to micrometer conversion (assuming a fixed pixel for an optic disc diameter for a population and average optic-disc diameter of 1800 μm and/or 1850 μm). <u>Additional method:</u> use a personalized disc diameter assessment (i.e., disc diameter is assumed 1800 or 1850 μm , and the width of each optic disc calculated per eye is used to determine an 'individualized' conversion factor).	Evaluation of the impact of generalized and personalized conversion methods using disc size as a reference in large epidemiological studies. The use of disc-fovea distance as a reference may also be considered. Development of conversion factors for all regions of the (peripheral) retina
5. Image quality	Use of standardized image quality assessment criteria specifically developed for visualizing the retinal microvasculature	Development of measure-specific image quality criteria. Insight can be acquired using simulation studies. The relevance and impact of the use of image restoration techniques for analyzing research questions in large-scale studies need to be established
6. Statistical analyses	<u>Selection bias</u> Acquire insight into the contributions of ocular comorbidities by conducting main analyses in the entire population and separate additional analyses with 1) adjustment for ocular comorbidities and 2) exclusion of eyes with ocular comorbidities. Consider the use of inverse probability weighting to reduce the impact of selection bias <u>Confounding bias</u> Use a minimum set of potential confounders (sociodemographic factors and factors included in Life's essential eight score). In a separate analysis, also evaluate the impact of potential confounders for which the evidence on their role as confounders is currently relatively less robust, but for which it is biologically plausible that they are confounders (e.g., air pollution, genetics, arterial stiffness, infections). Conduct analyses in which eye anatomy-related factors are considered (axial length, spherical equivalent, corneal measures, other). Examine differences in results according to key sociodemographic factors by testing for (statistical) interaction by age, sex, education level, and ethnicity <u>Information bias</u> Use a composite score that combines multiple retinal microvascular measures, or quantify individual retinal microvascular measures using results obtained from multiple photos per eye	<u>Selection bias</u> Report inter (and intra-) observer reliability per measure using optimal processing steps <u>Confounding bias</u> Further identification of potential confounders and acquisition of more robust evidence for less-well-studied factors <u>Information bias</u> Examine whether the use of a composite score yields more accurate results than analyses using individual microvascular measures. Determine optimal methods for selection of the region of interest (either using magnification factor correction or adjustment for eye anatomy-related factors). Quantify the increase in accuracy of retinal microvascular measures when retinal microvascular measures are extracted from multiple fundus photos

Table 2 provides an overview of recommendations for the standardization of the quantification of retinal microvascular imaging features and conducting statistical analyses.

Abbreviations: DR, diabetic retinopathy; AMD, age-related macular degeneration; DD, disc diameter; CRAE, central retinal arteriolar equivalent; CRVE, central retinal venular equivalent.

datasets available, manually annotated vessel maps from certain cameras and certain populations may not be available (Khan et al., 2021). If a researcher would like to work with images from a certain camera or population on which analysis tools have not yet been validated (e.g., a novel type of camera), we recommend that the researchers annotate a subset of images and compare the performance of several (open source)

analysis tools on images acquired specifically with this camera and/or in this population.

Fifteen images for each type of image (fovea- or optic-disc, thus 30 images in total) could be considered as a sufficiently large number for obtaining a valid result in a specified population (e.g., a population of a certain ethnicity that is free of eye diseases). Annotation of arterioles

and venules on one 45-degree fundus photo may take around 4-6 h, and on one (ultra)widefield image, around 18-20 h. Further research is warranted to provide evidence-based insight into the optimal number of images that require annotation for validation of a tool.

5.2. Anatomic landmark detection

We recommend reporting the performance of the algorithm for optic disc location detection separately per camera and per study population (and use standardized image grading quality). The result would ideally be expressed as a Dice segmentation score, and a manually annotated image by an expert grader would be used as ground truth. For validation of optic disc position detection, for example, 100 images (50 fovea-centered and optic-disc-centered photos) could be annotated. Annotation of an optic disc may take around 5-10 min per image. Further research is warranted to provide an evidence-based conclusion on the optimal number of images that require annotation to validate a tool.

Fovea location could also be used as an anatomic position and could be used to determine a standardized region of interest on fovea-centered images. Accuracy for fovea detection would in this case, also require validation for each camera and population. Performance of the algorithm for fovea detection could be expressed as the Euclidean distance in pixels between the center of the fovea detected by the algorithm and the manual annotator, as proposed by researchers who developed VascX (Vargas Quiros et al., 2025).

5.3. Region of interest

The exact vessels imaged differ for images with different fields of view (i.e., optic disc-centered, fovea-centered, and (ultra)widefield images). Hence, the region of interest also differs between these types of photos.

5.3.1. Optic disc-centered images

For optic disc-centered photos, there is a general agreement throughout the literature that zones A, B, and C are valid to use as standardized regions for the calculation of quantitative retinal microvascular imaging features. Zone A is defined as the zone between 0 and 0.5-disc diameter around the optic disc; zone B is defined as the zone between 0.5 and 1.0-disc diameter around the optic disc; and zone C is defined as the zone between 0.5 and 2.0-disc diameter around the optic disc (Fig. 3). Indeed, ten tools used one or more of these standardized zones. In our opinion, the use of zone C could be recommended over zones A and B as zone C is larger and may potentially be more informative and more reliable (i.e., robust to measurement error). Indeed, the test-retest reliability of SIVA-DLS for the calculation of CRAE and CRVE from Zone C across ten datasets was generally somewhat higher than when calculated for Zone B (Cheung et al., 2021).

5.3.2. Fovea-centered images

For fovea-centered photos, several regions of interest have been proposed (Fig. 5). However, there is currently no clear consensus on which is the most optimal region of interest to use in epidemiological studies. Further research is needed to identify whether there is an optimal region of interest that is most reliable and informative. The optimal exact region of interest would be a region that is detectable for the majority of individuals. The exact size of this region may depend on characteristics of the study population and could be specific to the exact measure under study. In a study among patients with high myopia, a relatively smaller region of interest could be chosen, as a smaller part of the retina is visible in eyes with a longer axial length in phakic eyes than in eyes without myopia. In a population study a larger region of interest could be chosen as high or pathological myopia is relatively less common in this kind of population.

Different approaches could be used to identify the region of interest accurately. One method could be to combine optic disc detection with

magnification factor assessment (Pors et al., 2024). A second method could be to use multiple anatomic landmarks to determine a standardized region of interest (e.g., optic disc and fovea location and/or fovea-optic disc angle) (Vargas Quiros et al., 2025).

5.3.3. (Ultra)widefield images

For (ultra)widefield photos, several regions of interest have been proposed. There is no clear consensus across the literature on the optimal region of interest. Further research in which multiple regions of interest are compared is needed to determine an optimal region of interest. Ideally, the largest possible region of interest should be identified that is detectable across (the majority of) individuals. This could be a region similar to the region proposed by the researchers who developed the VAMPIRE analysis tool (an ellipsoid shape of $120^\circ \times 85^\circ$, with the optic disc at its center) (Ma et al., 2024). Additionally, the choice of region of interest may also affect the choice of retinal microvascular measures.

As retinal microvascular dysfunction is considered a generalized phenomenon that (similarly) affects the microvasculature throughout the entire retina (Stehouwer, 2018), it could be assumed valid to combine results from individual vessels into a retinal microvascular composite score (as is also done for the calculation of CRAE and CRVE, where the widths of six vessels are combined into one value) (Knudtson et al., 2003). However, there are several reasons why it could also be relevant to separately analyze specific regions of interest. One, there are regional differences in the anatomy of the retinal periphery (Quinn et al., 2019; Bek, 2013). Retinal ganglion cells are three times more frequently identified in the nasal quadrant peripheral of the macula than in the temporal quadrant, the nasal quadrant may be more densely covered with retinal microvascular vessels (Quinn et al., 2019). Two, there may be differences in the sensitivity of certain peripheral regions to pathological changes. Indeed, signs of diabetic retinopathy are thought to be detectable earlier in the superior part of the peripheral retina than in other peripheral parts of the retina (Bek, 2013). Regional differences may also occur in other diseases such as Alzheimer's disease (e.g., some studies suggest amyloid beta is found most abundantly in the superior and inferior parts of the retina) (Gaire et al., 2024). Three, there may be differences in the accuracy of quantification of the retinal microvasculature due to technical reasons. For example, distortions of image quality on images acquired with Optos images are thought to occur more frequently in the nasal and temporal parts of the retina than in the superior and inferior parts of the retina (Quinn et al., 2019). Also, there are quadrants that are more affected by imaging artefacts than others, such as the infero-nasal quadrant, in which eyelids and eyelashes are relatively more common. Hence, future studies are warranted to compare the impact of using different regions of interest. The presentation of results for multiple regions of interest would be most informative and may be considered. The use of methods to account for magnification errors (through eye anatomy and camera setup related factors) and/or multiple anatomic landmarks may allow for the accurate determination of a standardized region of interest visible in all eyes (Pors et al., 2024).

Last, for all types of images (but in particular for (ultra)widefield images), the accuracy of the region of interest of detection may be further improved if the spherical shape of the eye is accounted for when analyzing images. The majority of tools do not consider the curvature of the retina. Methods for accounting for the spherical shape of the retina are in development (van den Berg et al., 2025).

5.4. Mathematical formulae

There is considerable variation between tools in the selection of quantitative retinal microvascular measures for calculation. The majority of tools calculate CRAE, CRVE, AVR (Knudtson et al., 2003), but other measures are not calculated by all tools (e.g., certain fractal dimension measures (D_1 , D_2 , singularity length) or relatively novel

measures such as alpha shapes) or are not available at all (e.g., vasculature curvature index). Most tools use the same methods to calculate the measures, but there is some variation between tools.

To ensure comparability across studies, we recommend that future studies report a certain minimal set of measures. A minimal set may contain one or more measures of different categories (e.g., some width measures (CRAE, CRVE, AVR), a fractal dimension measure (D0), a tortuosity measure (simple tortuosity), and a branching angle measure (mean branching angle). Ideally, an international consensus would be developed to determine which variables to use. This choice should be informed by insight into which features are most robust to measurement error and sensitive to detecting changes in microvascular health. Some measures have already been compared, and the results of these comparisons could be used to select measures (Chen et al., 2020). Validity of measures has been examined using several approaches, including against visual assessment of tortuosity by clinicians and for discrimination between eyes with and without diseases. For tortuosity, findings suggest that other measures than simple tortuosity could be recommended (Chen et al., 2020). Of note, it is important to consider the region of interest when developing these recommendations, as in certain situations it may not be valid to use measures across all types of photos. For certain fractal dimension measures, results from optic disc and fovea-centered photos may be comparable, and the exact region of interest may be less relevant, but for certain width measures (e.g., CRAE and CRVE) there may be considerable differences between results calculated from fovea or optic disc-centered photos.

Further research is needed to evaluate which mathematical formulas or methods for vessel calculation are optimal in situations where existing (open-source) tools disagree, such as in branching angle calculation. Performance of measures to discriminate between healthy and diseased eyes and interobserver test-retest reliability, preferably expressed as an intraclass correlation coefficient, could be used as measures to evaluate this (Ahlbom, 2021). Additionally, the use of Bland-Altman plots may also be considered, as this allows for more visual interpretation of the differences to facilitate better understanding (Ahlbom, 2021). Also, it is important to standardize how to take into account the anatomic variation of the microvasculature when calculating metrics. For example, the calculation of CRAE and CRVE from optic disc-centered images is done using the six largest arterioles or venules. But in those eyes in which there are less than six arterioles or venules visible (due to anatomic variation), CRAE and CRVE need to be calculated in a manner so that diameters are comparable between eyes. Moreover, this should be done in the same manner across studies (Tuma et al., 2025).

Future studies may also seek to develop novel quantitative retinal microvascular measures or combine multiple quantitative retinal microvascular measures into composite scores. Novel quantitative retinal measures may be developed using machine-learning-based methods. Composite scores may be developed that combine multiple quantitative retinal microvascular measures that assess a similar construct. Such a composite score may be more robust to measurement error than individual measures (Ahlbom, 2021).

5.5. Pixel to micrometer conversion

For central retinal images, the most precise method for converting pixels to micrometers may be a 'personalized' conversion method. This would mean that for each eye, the conversion factor is determined based on optic disc diameter measurement, rather than a group-level average conversion factor (Schanner et al., 2023). However, whether differences in conversion results lead to relevant differences in population-based studies, and whether these differences persist after adjustment for potential confounders, remain unknown (Schanner et al., 2023). Further, in the case of pathological changes of the optic disc, a personalized disc assessment could lead to a relatively less accurate and spurious estimation of retinal microvascular measures. Alternative options may be used to for pixel to μm conversion, e.g. other retinal anatomic landmarks

could be used (such as fovea-optic disc distance and angle), or this could be done using additional information on ocular anatomy (e.g. corneal thickness, axial length) and optical properties of the fundus camera used. Future large-scale studies are warranted to determine whether these methods yield relevant differences in findings. Furthermore, evaluation of methods that use the disc-fovea distance rather than the disc diameter as an anatomic landmark for determining pixel-to-micrometer conversion is also warranted.

In the retinal periphery, it may be important to use conversion factors that are specific to the exact location of the retina where measurements are conducted. Due to the spherical shape of the eye, the distances between the camera and retina are not equal at all locations of the retina (Pellegrini et al., 2014). If differences in conversion factors per region of the retina are not accounted for, differences in width measurements between the central and peripheral retina may be underestimated (Ahlbom, 2021; Pellegrini et al., 2014).

5.6. Image quality

5.6.1. Microvasculature

There are no widely accepted image quality criteria specific to grading the visibility of the retinal microvasculature, therefore we recommend that one is to be developed for this. This could be one by conducting simulation studies in which the exact impact of errors in image quality on microvascular measures is examined (Eid et al., 2025; Shi et al., 2022). Minimal sources of error that need to be considered are: the presence of artefacts that obscure partially or entirely the chosen region of interest, poor illumination, blurriness, and errors in image centering (Li and Chen, 2023).

Further research is also warranted to gain insight into the extent to which errors in image quality can be reversed by applying image restoration techniques. Image restoration techniques use mathematical approaches to digitally invert the progress of image degradation and can be used to correct for errors in illumination, color, haziness, and contrast enhancement (Zhang and Berendschot, 2024). Image restoration techniques have shown potential value in small studies, but whether there is value in using image restoration techniques in the different study designs warrants further investigation (Zhang and Berendschot, 2024).

5.6.2. Anatomic landmark

Standardized image quality criteria specific for the assessment of accuracy for the detection of optic disc (and possibly fovea) position are warranted. One option could be to use the FundusQ-Net score, which is specifically developed for visualizing the optic disc and peripapillary area. Other scores may also be considered (Yang et al., 2025).

5.7. Statistical analyses

Observational studies are prone to three types of bias, i.e., selection bias, confounding bias, and information bias (i.e., measurement error). Below, we explain these sources of bias and provide recommendations on how to standardize statistical analyses to minimize the impact of these types of bias.

5.7.1. General explication on sources of bias

Selection bias implies that the results found in a study population are not representative of the target population. For example, the quality of collected data may frequently be of poorer quality among individuals with worse health than among those in better health. This may lead to the selection of a population that is relatively healthier than the original target population (Ahlbom, 2021). This may lead to an underestimation of the associations between risk factors and disease outcomes (Hutcheon et al., 2010). Next, unequal distribution of potential confounders across populations may bias the estimation of the strength and direction of associations under investigation. Lack of adjustment for potential confounders may result in over- or underestimation (Ahlbom, 2021).

Information bias concerns bias in the measurement of the phenotype under study (e.g., retinal microvasculature). Information bias may lead to underestimation of the strength of the associations under study ("regression dilution bias") or reduce statistical power (Hutcheon et al., 2010).

5.7.2. Selection bias

To reduce the risk of selection bias, we recommend conducting the main analyses on the largest possible study population and, in additional analyses, assessing the robustness of these findings when applying stricter selection criteria (Ahlbom, 2021; Kashani et al., 2025). For example, in the main analyses we recommend including all individuals with data available and when applying minimal image quality criteria (Kashani et al., 2025); and in additional analyses, we recommend repeating analyses when excluding individuals with eye diseases and when using more strict image quality criteria (Ashraf et al., 2021). This allows us to assess the impact of the presence of eye diseases and image quality on results. To maximize the sample for analyses, we recommend using statistical methods to account for missing data (e.g., multiple imputation of missing covariates (Ahlbom, 2021) or inverse probability weighting (the latter is a method that uses weights to adjust the betas obtained in statistical analyses)) (Chesnaye et al., 2022).

5.7.3. Confounding bias

We recommend considering the largest possible number of potential confounders in statistical analyses to minimize the impact of confounding bias. Confounders that are recommended to be considered include sociodemographic risk factors (age, sex, education level, ethnicity) and modifiable risk factors (blood pressure, blood pressure-lowering medication, cholesterol, cholesterol-lowering medication, diabetes, smoking, diet, sleep, physical activity, and overweight) (Cheung et al., 2022a; Lee et al., 2025; Li et al., 2016a; Zeng et al., 2025). The latter are factors that are part of the Life's Essential Eight score, which has been developed by the American Heart Association (Lloyd-Jones et al., 2022). All these factors are associated with quantitative retinal microvascular measures (Lee et al., 2025). Further, other potential confounders may be considered, such as air pollution (Millen et al., 2024), indoor environment (van Dyck et al., 2025), genetic predisposition to microvascular dysfunction (Zekavat et al., 2022; Tomasoni et al., 2023; Jiang et al., 2023; Jensen et al., 2016; Yu et al., 2025), arterial stiffness (Liu et al., 2019), and infections (e.g. COVID-19, tuberculosis) (Cunha Filho et al., 2024; Basu et al., 2025; Fitt et al., 2021; Boillot et al., 2015). The evidence on the role of these variables as potential confounders is less strong than for other confounders, but it is biologically plausible that these factors may be potential confounders. Although the impact of adjustment for each factor may be small, adjustment for a combination of these factors may have a relevant impact on effect estimates (Ahlbom, 2021).

In further analyses, the impact of variables that may in part be potential confounders and in part be mediators can be considered (Zhu et al., 2025a; Schisterman et al., 2009). For example, in the association between quantitative retinal microvascular measures and chronic kidney disease, certain factors such as cardiovascular disease (Cheung et al., 2021; Girach et al., 2024; Zhang et al., 2024b), and systemic inflammation (Liu et al., 2021) may be confounding factors and/or mediators. We also recommend conducting analyses in which eye anatomy-related factors are accounted for, such as spherical equivalent, axial length, and corneal biomechanics measures (such as corneal thickness and curvature) (Pors et al., 2024). Other ocular measures may also be considered, such as intraocular pressure (Arciero et al., 2013).

We further recommend separate analyses in which the impact of ocular comorbidities is examined (Kashani et al., 2025). We recommend conducting a separate analysis in which associations are adjusted for key ocular comorbidities (i.e., glaucoma (O'Neil et al., 2025; Wang et al., 2007; Chen et al., 2025), diabetic or hypertensive retinopathy (Cheung et al., 2012, 2022a), and myopia (Lin et al., 2026)) (Kashani et al.,

2025). All these diseases have been associated with quantitative retinal microvascular imaging features (Cheung et al., 2022a; Cheung et al., 2012; O'Neil et al., 2025; Wang et al., 2007; Chen et al., 2025). Age-related macular degeneration may also be considered, but most of the studies on the association between quantitative retinal microvascular measures and age-related macular degeneration did not consistently find that age-related macular degeneration is a determinant of quantitative retinal microvascular imaging features (Toulouie et al., 2022). Next, we recommend conducting additional analyses excluding individuals with ocular comorbidities (Kashani et al., 2025). On the one hand, this may reduce the impact of measurement error arising from changes in eye anatomy in certain ocular diseases (Zhu et al., 2025a). On the other hand, this may increase the chance of selection bias (Ahlbom, 2021).

Researchers may further seek to obtain mechanistic insight into whether arteriolar or venular width drives associations between systemic risk factors or diseases and quantitative retinal microvascular width measures (e.g., CRAE). Such insight can be obtained by examining whether associations change in strength when both arteriolar and venular width measures are entered into the same model compared with only one of these measures. If the association observed for one retinal measure attenuates after entering both width measurements in the model, this may indicate that the association is, at least in part, driven by the width of the other vessel (Gould et al., 2017). This analysis is currently not systematically conducted throughout the literature.

Further, we recommend conducting statistical tests for interactions to determine whether associations may differ in strength according to sociodemographic factors, such as age, sex, educational level, ethnicity or other factors (Ahlbom, 2021). Tests for interaction by diabetes and hypertension can be considered biologically plausible, as certain associations may be different under hyperglycemic or hypertensive circumstances (Stehouwer, 2018). Under hyperglycemic circumstances, the effects of certain risk factors, especially those that increase hemodynamic stress, may be stronger as hyperglycemia impairs autorregulation (Tapp et al., 2022). Likewise, under hypertensive versus normotensive circumstances, already lower levels of microvascular dysfunction may predispose to capillary hypertension, rupture, and subsequently ischemia and tissue damage (van der Heide et al., 2022; Stehouwer, 2018).

5.7.4. Information bias

The impact of information may be minimized if optimal choices are made when selecting tools for vessel segmentation and arteriole and venule classification; anatomic landmark detection, region of interest selection; selection and calculation of quantitative retinal microvascular measures; conversion of pixels to micrometers; and image quality assessment (Table 2).

Information bias may further be reduced by calculating quantitative retinal microvascular features using multiple measures combined in a composite score (Zhu et al., 2025a). Composite score can be used to combine measures that reflect a similar underlying construct into one variable. This is done by averaging individual variables and is thought to yield a more stable measure that are less sensitive to measurement error than when analyzing quantitative retinal microvascular measures individually. Another approach is to collect multiple images for each eye and determine the average value for each measure per eye. It has been suggested to acquire three images per eye and average the results of these measurements (Hanssen et al., 2022), but it remains unknown whether this approach yields a relevant benefit in the accuracy of quantitative imaging features in (large-scale) studies. Further research is warranted to evaluate this. Such studies may also seek to identify the optimal number of images required to reduce measurement error significantly. Ideally, such studies should provide quantitative insight into the added value for each additional image and specific for each measure. Last, researchers may also combine the results of both eyes per measure by averaging the results. This is a commonly used approach,

but it is not standard throughout the literature (Schneider et al., 2025).

Researchers may present results for each eye separately and consider the use of statistical models that combine microvascular measures from multiple images and eyes. The selection of statistical methods to account for strongly correlated variables is relevant when analyzing multiple variables (Chun and Keles, 2010). Inability of the model to discriminate between highly correlated measures due to insufficient statistical power and/or strongly correlated variables (i.e., multicollinearity) may lead to biased results of statistical models (Ahlbom, 2021).

Future studies may also seek to compare whether detection of the region of interest or adjustment for eye anatomy-related variables yields different results in (large-scale) studies. The region of interest can be determined by calculating the magnification factor of the eye from eye anatomy and camera setup-related variables (Pors et al., 2024). Alternatively, it may be possible to account for variation in the region of interest by adjusting for eye anatomy-related variables in statistical analyses (e.g., axial length, spherical equivalent, and corneal biomechanics-related variables). Both approaches may yield similar results, but formal comparisons of the methods are currently unavailable. Studies that seek to evaluate this should consider whether methods that select a region of interest based on anatomic measures perform equally well in phakic and pseudophakic eyes. If the use of eye anatomy-related variables yields more accurate results, these measurements may be included in recommendations for the design of future epidemiological studies. When optimal processing steps have been determined, it is important to report the results of inter- and intra-observer reliability. We recommend including at least 50 images in such evaluations. Results of these evaluations are recommended to be expressed as an intraclass correlation coefficient.

The impact of using dilating drops on quantitative retinal microvascular measures has been examined in a few studies (Cirla et al., 2023; Frost et al., 2019). Overall, these studies suggest that dilating drops may have a small impact on the visualization of the retinal microvasculature, but do not significantly affect the results of statistical analyses (Frost et al., 2019).

6. Discussion

The present review has two main contributions. First, we conducted a comprehensive comparison of commonly used tools for quantifying retinal microvascular features (retinal vasculometry) from both standard color and (ultra)widefield fundus images. This comparison revealed considerable variation across tools in the reporting, validation, and performance of multiple key steps in quantitative retinal microvascular feature analysis. Second, we proposed a methodological framework for standardizing retinal microvascular imaging feature quantification and for conducting statistical analyses.

This review fills an important gap in the literature. Previous studies have compared commonly used tools and identified differences (Eid et al., 2025; Mautuit et al., 2022; Yip et al., 2016; French and Heitmar, 2021), but they have not systematically examined how these differences arise at each step of the quantification process for retinal microvascular imaging features (Chew et al., 2025; Mahapatra et al., 2024). Other studies have compared measures and shown that image quality affects quantitative retinal microvascular measures (Shi et al., 2022; Deepika et al., 2021), but no studies have proposed a framework for standardizing retinal microvascular imaging feature quantification or provided detailed recommendations for statistical analysis.

6.1. Current evidence

Over the last 20 years, many studies worldwide have investigated quantitative retinal microvascular imaging features. Initial studies from the USA (Beaver Dam Eye Study) and Australia (Blue Mountain Eye Study) (Cheung et al., 2022a; Hanssen et al., 2022) were followed by numerous population-based studies conducted on all continents among

individuals at all stages of life (Li et al., 2016a). The number of ocular and systemic diseases in which quantitative retinal microvascular imaging features have been studied has importantly increased. The current body of evidence is dominated by studies that examined the role of these features in diabetic retinopathy (Perais et al., 2023), cardiovascular disease (Cheung et al., 2021; Girach et al., 2024; Zhang et al., 2024b), diabetes (Blindbaek et al., 2017), and hypertension (Cheung et al., 2022a). But increasingly evidence is also being acquired for other ocular diseases (e.g., glaucoma (O'Neil et al., 2025; Wang et al., 2007; Chen et al., 2025) and amblyopia (Wagner et al., 2024)) and other systemic diseases (e.g., Alzheimer's disease and related dementias (Ashraf et al., 2023), Parkinson's disease (Ma et al., 2024; Vlasov et al., 2017), late-life depression (Kennedy et al., 2023), schizophrenia (Kennedy et al., 2023), multiple sclerosis (Jiang et al., 2024), and bipolar disorder (Kennedy et al., 2023)). Also, increasingly more systemic determinants of retinal microvascular health have been determined, and the bidirectional interactions between microvascular health and systemic risk factors such as hypertension and diabetes have been unraveled (Stehouwer, 2018).

The most recent significant advances in research on retinal microvascular imaging features come from studies that use genetic instruments to assess retinal microvascular and/or systemic health (i.e., polygenic risk scores or Mendelian Randomization studies) (Zekavat et al., 2022; Blose et al., 2024; Arnould et al., 2025). Indeed, genetic studies have replicated many findings from observational studies (Zekavat et al., 2022; Blose et al., 2024; Arnould et al., 2025; Li et al., 2016b). These studies add value to results from observational studies, as they are less susceptible to bias from unmeasured confounding than observational studies (as genetic predisposition is determined at conception and is not influenced by environmental factors over life) (Yeung et al., 2025).

Although the current body of evidence has numerous important strengths, there are still certain major limitations in our knowledge. Relatively few studies have examined the temporal and reversible nature of changes in quantitative retinal microvascular imaging features (Lee et al., 2025; Kaur et al., 2022; Pead et al., 2021). As such, it remains incompletely understood *when* which retinal microvascular changes occur and *to what extent* these changes can be *reversed* via interventions. Some studies have shown that modification of certain risk factors is associated with changes in quantitative retinal microvascular imaging features (e.g., use of antihypertensive (Ozcimen et al., 2022) or cholesterol-lowering medication (Sasaki et al., 2013), weight loss (Marta et al., 2023), modification of an unhealthy diet (Repo et al., 2024), and interventions such as carotid endarterectomy (Ala-Kauhala et al., 2023) and intravitreal injections (Mitsch et al., 2020)). However, these studies were generally relatively small. It remains unclear whether the identified associations are valid in the general population. Further, there are relatively few studies that have compared microvascular dysfunction between the central and peripheral retina. Preliminary findings from small-scale studies suggest that retinal microvasculature in the periphery could be more susceptible to deterioration than the microvasculature in the central retina (Ashraf et al., 2021; Csincsik et al., 2018; Leung et al., 2024; Pead et al., 2023). Thus, peripheral retinal microvascular imaging features could have an added value to central retinal microvascular imaging features. Although a strength of peripheral retinal imaging is that more retinal vessels can be imaged, a potential limitation is that the accuracy with which vessels can be visualized is lower than in the central retina (Quinn et al., 2019).

6.2. Value proposition of retinal microvascular imaging features

Retinal vasculometry provides an important opportunity to understand better the role of microvascular dysfunction in ocular and systemic disease. However, the clinical value of retinal microvascular imaging by itself for risk stratification and disease classification seems limited. The predictive value of retinal microvascular imaging features is considerably less than the predictive value of imaging features developed using

medical AI systems (Zhu et al., 2025a). For example, AI-based analyses of color fundus photos predict incident stroke (Jiang et al., 2025) with much higher accuracy than quantitative retinal microvascular imaging features (Rudnicka et al., 2022; Cheung et al., 2022b). Likewise, similar results have been obtained for other diseases such as Alzheimer's disease (Cheung et al., 2022b; de Jong et al., 2011; Correction to, 2024). However, although quantitative retinal microvascular imaging features by themselves may not be scalable features for use in the clinic, they could contribute to the development of scalable imaging features that can make an impact in the clinic. One, the development of features from a combination of retinal microvascular imaging features and unsupervised AI methods may yield features with a better predictive performance than when only unsupervised methods are used (Saito et al., 2025). Two, quantitative retinal microvascular measures may be used to enhance insight into the explainability of AI-based algorithms, which may improve the willingness of clinicians to use retinal imaging features in clinical practice (Zhu et al., 2025a).

6.3. Future research

Future research may advance the field by improving research methodology, developing clinical tools that incorporate retinal microvascular imaging features, and deepening understanding of the role of microvascular dysfunction in ocular and systemic disease. Recommendations per topic are outlined below.

6.3.1. Methodological

We recommend conducting detailed comparisons and validating open-source tools, and selecting optimal algorithms from open-source systems, specific to cameras and the population under study. Parts of different processing pipelines could be combined. Next, more research is needed to identify optimal regions of interest, select a minimal set of quantitative retinal microvascular measures, compare measures (e.g., fractal dimension measures), define standardized image quality criteria, and determine methods for pixel-to-micrometers conversion. Work on these topics is being undertaken by researchers within several international consortia, including the European Eye Epidemiology consortium and the professional interest area 'Eye as a biomarker for Alzheimer's disease' from the Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART) (Kashani et al., 2025). Frank and Tos actively contribute to this (Kashani et al., 2025).

Further, there is a need for more available, open-source datasets with accurate vessel segmentation annotations, made according to universally accepted methods. Ideally, such datasets should be developed using various cameras, in multiple ethnic populations, for individuals with and without (common) eye diseases (such as diabetic retinopathy, age-related macular degeneration, (pathological) myopia, and glaucoma).

6.3.2. Clinical tools

Future studies may seek to develop multimodal retinal imaging-based prediction models that incorporate color or (ultra)widefield fundus imaging (Chan et al., 2024). A combination of fundus and optical coherence tomography angiography imaging may be promising for developing microvascular biomarkers, as it allows visualization of all levels of the retinal microvasculature. Further studies are needed to gain better insight into the temporality and reversibility of retinal microvascular changes (Drachmann et al., 2025; Han, 2012). If changes in quantitative retinal microvascular measures occur in an early phase or before disease onset and are reversible by the given interventions, these measures could serve as tools for monitoring disease progression and/or responsiveness to treatment (Drachmann et al., 2025; Han, 2012). Future studies may also seek to identify which retinal microvascular measures are most promising for risk stratification. Comparison between microvascular measures would be facilitated if studies would report results per measure in units that can be easily be compared, e.g. as

z-scores (i.e. expressed in standard deviations). Some studies compared findings for different retinal measures (e.g. diameters, fractal dimension, tortuosity) and found that associations of individual retinal measures with key clinical conditions were similar in strength (stroke (Yusufu et al., 2025b), hypertension (Betzler et al., 2022), major cardiovascular adverse events (MACE) (Mordi et al., 2022) and Alzheimer's disease (Ashraf et al., 2023)).

Future studies are also warranted to determine the exact requirements needed for biomarkers to have a clinical impact (Zhu et al., 2025a). Ideally, needs would be accurately specified for sensitivity, specificity, positive and negative predictive values and should be supported by cost-effectiveness studies. For certain specific diseases and clinical settings, such recommendations have already been developed. For example, for Alzheimer's disease, there is a consensus on which performance biomarkers should be used in primary and secondary care settings to be useful (Schindler et al., 2024). However, for many diseases, this is currently less well-defined. After this, subsequent studies are warranted to determine how ocular biomarkers could be integrated within existing healthcare pathways.

6.3.3. Mechanistic and etiological studies

There are still many gaps in our understanding of the mechanisms that underlie how microvascular dysfunction contributes to ocular and systemic disease. Future studies are warranted to provide better insight into 1) the contribution of neurodegeneration and microvascular structures to dysfunction of the neurovascular coupling unit (van der Heide et al., 2022); 2) the interplay between microvascular dysfunction at different levels of the retinal vasculature (e.g. between terminal and precapillary arterioles) (Jensen et al., 2019); 3) the interaction between microvascular dysfunction and other disease mechanisms (e.g. amyloid beta accumulation in Alzheimer's disease) (Steinman et al., 2020); 4) whether the pathobiology of microvascular dysfunction in ocular and systemic diseases differs between the central and peripheral parts of the retina (Quinn et al., 2019; Bek, 2013); and 5) paradoxical findings that remain incompletely understood (for example, the mechanisms underlying arteriolar widening (instead of arteriolar narrowing) in certain conditions such as late-life depression) (van Gennip et al., 2024). It is important that the systemic disease context of the patient is taken into account when selecting and interpreting microvascular features. For example, the pathophysiological effects of hypertension, hypoxia, and hyperglycemia on the retinal microvasculature are different. Hypertension is thought to induce arteriolar narrowing, but hypoxia and hyperglycemia are thought to induce arteriolar widening. The systemic context may be less relevant for the interpretation of other microvascular measures, e.g. both hyperglycemia and hypertension are associated with lower arteriolar and venular fractal dimension and greater arteriolar and venular tortuosity.

Recent technological developments have enabled new modalities for visualizing and quantifying the retinal microvasculature, including adaptive optics imaging and OCT-A (Chua et al., 2024). With adaptive optics imaging, the retinal microvasculature can be imaged at an almost histological resolution, allowing accurate visualization of the arteriolar vessel wall separate from the blood column in the central retina (Burns et al., 2019). Adaptive optics imaging may deepen our understanding of the remodeling of the vessel wall (Burns et al., 2019). Advances in OCT-A imaging enable visualization and quantification of capillaries in the inner retina, both in the central and peripheral retina (Chua et al., 2024). Hence, a combination of widefield OCT-A and (ultra)widefield fundus photography can enable further insight into structural retinal microvascular changes and microvascular dysfunction in both central and peripheral retina (Chua et al., 2024). Furthermore, with novel OCT-A technologies (e.g., swept-source), choroidal vessels can be imaged with high resolution. Studies of choroidal vessels may further provide insight into microvascular dysfunction. There are further developments in methods for in vivo quantification of blood flow and oxygen levels, and these methods are increasingly accurate (e.g.,

Doppler OCT imaging compared with laser speckle flowgraphy) (Bohm et al., 2022). These methods can enable to better understand the consequences of structural microvascular changes on flow and the impact of oxygenation on retinal microvascular width measures (Bohm et al., 2022). One possible explanation for the paradoxical finding that arterioles are wider rather than narrower in certain diseases (e.g., late-life depression) is that systemic oxygen levels are lower (Totou et al., 2023; Liew et al., 2014; Choudhary et al., 2013). Multimodal studies that include oxygenation measurement could provide further insight into this. In addition, technological advances allow for the acquisition of higher-resolution or faster images, which may increase the accuracy of visualizing the retinal microvasculature (Chua et al., 2024). Lastly, the accuracy of biological models can further be improved by using three-dimensional analysis strategies (Dong et al., 2025; Narotamo et al., 2024).

6.4. Strengths and limitations

This review has several strengths. One, this review comprehensively evaluated the reporting and validation of a large number of commonly used tools. Two, we conducted a detailed comparison of open-source tools by examining their scripts and testing them. Three, the development of the framework for standardization was done in a multidisciplinary manner by researchers with backgrounds in relevant domains (i.e., ophthalmology, vascular biology, image processing, clinical translational research, and epidemiology).

This review also has certain limitations. One, we selected tools for comparison by conducting a comprehensive literature study, but did not use an exhaustive systematic search strategy. However, we believe that the main conclusions and the design of the proposed framework would likely not have been substantially different if a few more algorithms had been included. Two, we had access to all details of the processing for some analysis tools (i.e., open source tools VascX, AutoMorph, PVBM) but not for all analysis tools. This may have affected the results of our comparison. Three, the emphasis of this review was on standardizing processing steps to ensure comparable results across analytic tools. The exact steps are a multidisciplinary domain that involves both technical, biological, and medical expertise. The focus of this review was on the intersection of these fields, and some aspects were discussed in a more general way rather than in (exhaustive) detail. Future studies are warranted to in-depth examine and compare the state-of-the-art technical advances for all image processing steps. This will facilitate the selection of the best processing steps for the development of optimal future processing pipelines for future studies (Kipli et al., 2018). Four, because different tools were validated against different ground truths, direct numerical comparison of reported performance metrics across tools is not possible. Five, we did not discuss all available widefield fundus imaging cameras. We deem that the proposed framework can also be used to develop vessel analysis tools for other widefield cameras, not just those manufactured by Optos. We also did not evaluate OCT-A, which measures other features of the retinal microvasculature, particularly capillary dysfunction. Six, we did not discuss the relevance of standardizing key aspects of study design, outside of factors related to the quantification of microvascular imaging features and the conduct of statistical analyses. It is also important to standardize the definitions of outcomes and time intervals in clinical studies (Ahlbom, 2021). For example, in Alzheimer's disease, there may be an underestimation of effect sizes if individuals with a clinical diagnosis of Alzheimer's disease are included rather than biologically proven Alzheimer's disease (based on positron emission tomography imaging or cerebrospinal fluid biomarkers) (Jack et al., 2018). Indeed, the use of clinical diagnosis is associated with misclassification of Alzheimer's disease in up to 50% of patients (Jack et al., 2018).

7. Conclusions

Technological advances in image acquisition and processing over the last century have enabled non-invasive visualization of the retinal microvasculature in both the central and peripheral retina, as well as the availability of a plethora of quantitative measurements, including AI and deep learning. However, the full potential of this field (retinal vasculometry) is hindered by the limited comparability of existing analytic tools. In this review, we compared 15 commonly used tools and proposed a framework to standardize the quantification of retinal microvascular imaging features, thereby promoting valid, accurate, and comparable results across studies. This framework can help clinical researchers address key gaps in the literature, i.e., *when* retinal microvascular imaging features change over the course of disease (*"temporality"*), whether the microvasculature is *sensitive* to changes in systemic risk factors (*"reversibility"*), how microvascular dysfunction contributes to neurodegeneration, and how microvascular dysfunction differs between the *central* and *peripheral* retina. Clarifying these issues will deepen our understanding of the role of microvascular dysfunction in ocular and systemic disease and support the development of retinal vasculometry-based biomarkers using fundus photography for clinical disease prevention and mitigation.

CRediT authorship contribution statement

Frank C.T. van der Heide: Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Youba Zamoum:** Writing – original draft, Visualization, Investigation. **Mehdi Ounissi:** Writing – review & editing, Supervision. **Tos Tjm Berendschot:** Writing – review & editing, Methodology, Conceptualization. **Carol Y. Cheung:** Writing – review & editing. **Maya Koronyo-Hamaoui:** Writing – review & editing. **Leopold Schmetterer:** Writing – review & editing. **Jacqueline Chua:** Writing – review & editing. **Tunde Peto:** Writing – review & editing, Conceptualization. **Imre Lengyel:** Writing – review & editing, Conceptualization. **Lajos Csincsik:** Writing – review & editing. **Catherine Creuzot:** Writing – review & editing. **Louis Arnould:** Writing – review & editing. **Ingeborg Stalmans:** Writing – review & editing. **Emanuele Trucco:** Writing – review & editing. **Tom Macgillivray:** Writing – review & editing. **Anthony P. Khawaja:** Writing – review & editing. **Lisa Zhuoting Zhu:** Conceptualization. **Pearse A. Keane:** Writing – review & editing. **Tien Yin Wong:** Writing – review & editing. **Dan Milea:** Writing – review & editing, Supervision, Data curation, Conceptualization.

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